

ICMR AMRIT

Antimicrobial Resistance Intelligence Toolkit Application Manual and Standards Conformance Report

A complete operating manual for the local-first AMRIT desktop application, together with an evidence-based verification of its conformance to national and international health data terminology and interoperability standards.

Verification verdict

AMRIT is **substantially — but not fully — conformant**. Its structural interoperability is sound and its terminology catalogue is rich, but several standard codes that the application already stores are never emitted, and quantitative susceptibility results are exported without units. Chapter 21 states the position standard by standard; Chapter 22 lists twelve findings with reproducible evidence.

APPLICATION VERSION

2.0.0 (Electron, ARM64 macOS build)

DOCUMENT DATE

12 August 2026

REGRESSION SUITE

Typecheck + lint clean; 100 passed, 2 skipped

CATALOGUE VERSION

icmr-amrit-packaged-catalogue 2026.1

VERIFICATION BASIS

Compiled build + live database, 11 100 isolates

CLASSIFICATION

Internal — programme and site use

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Screens 1–29 are reproduced from the compiled 2.0.0 build running against a laboratory database of 11 100 isolates across five sites, driven over the Chrome DevTools Protocol on 12 August 2026. No screen in this manual is a mock-up, and no figure is decorative: each one is drawn from the source it describes.

1 · What AMRIT is, and what it is not

AMRIT is a local-first desktop application for antimicrobial-resistance surveillance. It holds the laboratory's own data on the laboratory's own machine, interprets susceptibility results against explicitly activated breakpoints, and releases information outward only through channels the operator chooses.

1.1 Scope

One installation supports many laboratories but works in exactly one at a time. Everything the application computes — every denominator, chart, export and federated answer — is scoped to the laboratory shown in the top bar. The application covers:

- Multi-laboratory configuration with a single active site.
- Guided isolate entry with a draft/final lifecycle, duplicate detection and retained AST results.
- Batch import from CSV, XLSX, XLSB and WHONET/BacLink-style files, with mapping, validation and atomic commit.
- Seventeen configurable master catalogues covering antibiotics, organisms, specimens, aliases, locations, domains, data fields, hospitals, states, districts, panels, breakpoints, QC ranges, expert rules, expected resistance, genomic markers and coded values.
- Versioned breakpoint sets with workbook staging, a provenance hash and explicit activation.
- Ten deterministic analysis modes and six export formats.
- Governed One Health capture across human, animal, environment, food, antimicrobial-use and infection-prevention workflows.
- Aggregate-only federation over long-poll or WebSocket.
- A searchable audit trail, and a separate hash-chained audit log for One Health.

1.2 Intended use and limits

Clinical and regulatory notice

AMRIT is a surveillance and laboratory-workflow tool. It is **not** autonomous clinical decision software and must not be used as one. Breakpoint content is edition-, method-, organism- and site-specific: a qualified laboratory authority must review the provenance of a staged breakpoint set and activate it deliberately before its interpretations are used. Paid breakpoint content is neither bundled nor redistributed. The application footer states, on every page, “For surveillance support — not autonomous clinical decision-making”.

1.3 The three principles the design defends

PRINCIPLE	HOW IT IS ENFORCED
Local-first	All patient-level data stays in a SQLite database inside the operating-system user-data directory. Nothing is uploaded implicitly; there is no background telemetry.
Aggregate-only federation	A node answers six named question types with counts, rates and FHIR MeasureReports. Two independent guards inspect every outbound payload and abort on any key or resource type that could carry identity.

PRINCIPLE	HOW IT IS ENFORCED
Evidence for every action	Saves, deletes, imports, breakpoint activations, exports and sync exchanges are written to the audit log with actor and outcome. One Health entries additionally form a SHA-256 hash chain.

1.4 What this manual covers

Parts I to III are the operating manual. Part IV is a standards-conformance verification carried out against this exact build: the packaged catalogue was measured field by field, real export artefacts were produced from the live database, and each finding cites the source location and the observed output that demonstrates it.

2 · Installation, first run and where data lives

2.1 Requirements

Runtime	Electron 43 desktop application; no server component required
Build toolchain	Node.js 24 or newer, pnpm 11 or newer
Platforms	macOS (dmg, zip), Windows (nsis, portable), Linux (Applmage, deb)
Database engine	SQLite through the Node <code>node:sqlite</code> module, WAL journal
Optional	A local Ollama service on loopback for the assistant

2.2 Building from source

```
pnpm install
pnpm run check      # TypeScript + ESLint, zero warnings tolerated
pnpm test           # Vitest regression suite
pnpm run build      # electron-vite build
pnpm run dist       # platform installers (runs check and test first)
```

`pnpm run dist:dir` produces an unpacked application without installers, which is what the verification in Part IV was run against.

2.3 First run

1. The application creates a private SQLite database under the platform user-data directory.
2. If the database is completely empty — no laboratory, isolate, import run or master content — the packaged catalogue is seeded once. Existing databases are never overwritten.
3. The packaged catalogue is validated before it is trusted: schema version, dataset name, version string, row counts, per-collection minimums, a canonical-JSON SHA-256 that must match the reviewed application manifest, referential integrity between panels and their organisms, specimens and antibiotics, and a scan that rejects any personally identifying key.
4. If `../desktop_app/whonet_replica.db` exists it is copied once and migrated in the private location. The legacy Python database is never opened for mutation.

Overrides for testing and validation

AMRIT_DATABASE_PATH nominates an explicit writable database file. AMRIT_LEGACY_DB nominates a legacy database for one-time copying. Both are intended for validation runs, so that a verification exercise never touches production data — this is how the screenshots and export artefacts in this manual were produced.

2.4 Where data lives

ARTEFACT	LOCATION AND HANDLING
amrit.sqlite3	User-data directory. Holds every patient-level record. Foreign keys on, WAL journal, 5-second busy timeout.
catalog-seed.v1.json	Packaged read-only resource, 3.4 MB, SHA-256 pinned in source.
genomic-markers.v1.json	Packaged read-only resource; 52 AMR determinants.

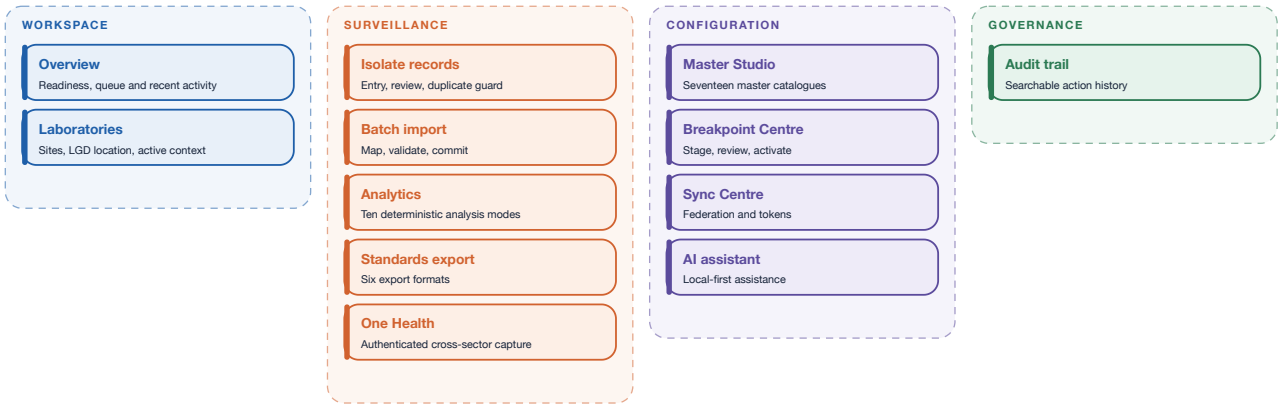
ARTEFACT	LOCATION AND HANDLING
Sync and site tokens	Encrypted with Electron <code>safeStorage</code> . If OS encryption is unavailable they stay in memory for the session and are never written as plaintext.
Exports	Written only to a path the operator selected in a save dialog. The renderer cannot supply a destination.

The absolute path of the active database is printed in the footer of every page, so an operator can always state which file produced a given number.

3 · The workspace

Twelve workspaces are grouped into four bands in the left rail. The top bar always shows the active laboratory and the federation status.

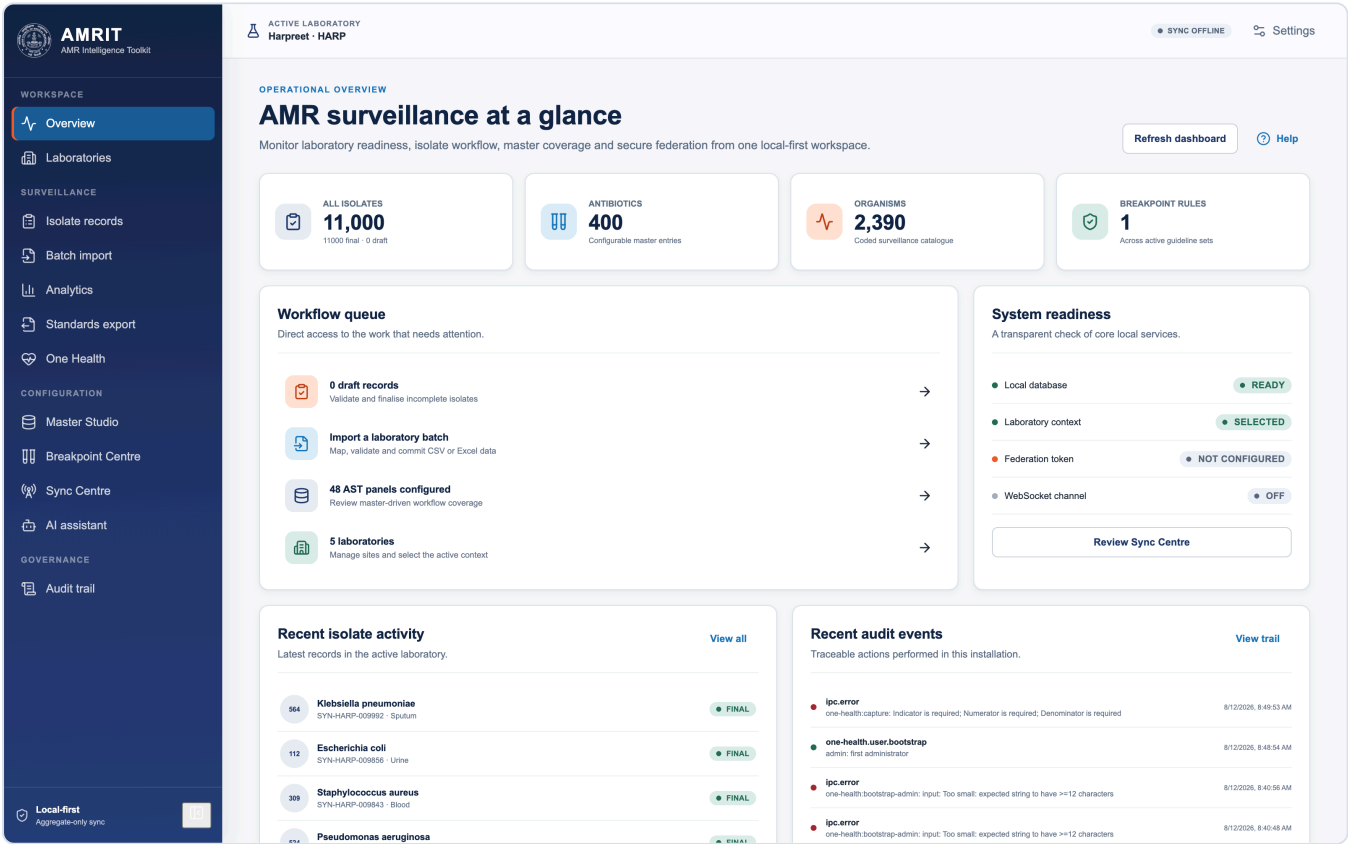
TWELVE WORKSPACES IN FOUR GROUPS



Persistent chrome

The left rail keeps the active workspace, a local-first badge and a collapse control. The top bar always shows the active laboratory and the federation status, because every figure, denominator and export in the application is scoped to that one laboratory. The footer of every page repeats the application version and the absolute path of the local database, so an operator can always state which build and which file produced a given number.

Figure 1 — Navigation map. Every workspace, its purpose and the persistent chrome that frames it.



Screen 1 — Overview. Counts for the active laboratory, a workflow queue that links directly to outstanding work, a system-readiness panel and the two most recent activity feeds. Federation shows **NOT CONFIGURED** until a token is supplied.

3.1 Reading the overview

PANEL	WHAT IT TELLS YOU
Counter row	Isolates split by final and draft, antibiotic and organism catalogue sizes, and the number of breakpoint rules across active guideline sets.
Workflow queue	Draft records awaiting completion, the import entry point, the number of configured AST panels and the laboratory count. Each row navigates to the work.
System readiness	Local database, laboratory context, federation token and WebSocket channel. A red or grey marker here explains why an action elsewhere is unavailable.
Recent isolate activity	The latest records in the active laboratory with organism, specimen number, specimen type and status.
Recent audit events	Traceable actions in this installation, including failures — the screen above shows genuine <code>ipc.error</code> entries from rejected One Health input, which is the audit trail working as intended.

3.2 Conventions used throughout the interface

- **Required fields** carry an asterisk and are enforced in the main process as well as the form.
- **Help drawers** open from the [Help](#) control on each page header.
- **Status pills** use one vocabulary everywhere: green for ready or final, orange for draft or attention, red for error, grey for inactive.
- **Destructive actions** — delete, disable, activate a breakpoint set — are always explicit and always audited.

4 · Laboratories and the active context

A laboratory record is more than a name: it carries the geography that national reporting depends on, and the interpretation defaults that decision support will apply.

AMRIT
 AMR Intelligence Toolkit

ACTIVE LABORATORY
Harpreet · HARP

● SYNC OFFLINE
⚙️ Settings

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SITE ADMINISTRATION

Laboratories

Register facilities and select the active laboratory context used by records, panels, imports, exports and federation.

[+ Add laboratory](#)
[? Help](#)

✔ Active site

Harpreet (HARP) is currently selected. Changing the active site never moves or merges existing records.

5 of 5 sites

CODE	LABORATORY	LOCATION	AST DEFAULTS	STATUS	ACTIONS
HARP	Harpreet ICMR Punjab validation	📍 Amritsar, Punjab, India	CLSI 2025 Disk diffusion	● ACTIVE	✓ Selected ✎ Edit 📄 Copy configuration 🗑
ICMR	ICMR Laboratory Domain not set	📍 New Delhi, Delhi, India	CLSI Disk diffusion	● AVAILABLE	Select ✎ Edit 📄 Copy configuration 🗑
ICMR_COPY	ICMR Laboratory (copy) Environment	📍 North Goa, Goa, India	EUCAST Gradient strip	● AVAILABLE	Select ✎ Edit 📄 Copy configuration 🗑
WF100	ICMR Workflow Validation Lab ICMR workflow validation	📍 Amritsar, Punjab, India	CLSI 2025 Disk diffusion	● AVAILABLE	Select ✎ Edit 📄 Copy configuration 🗑
WTH	WHO Tutorial Hospital2 Domain not set	📍 Faridabad, Haryana, World Health Organization	— Method not set	● AVAILABLE	Select ✎ Edit 📄 Copy configuration 🗑

ICMR AMRIT 2.0.0

Local database: /Users/hsingh/Library/Application Support/ICMR AMRIT Verification/data/amrit.sqlite3

For surveillance support—not autonomous clinical decision-making

🔒

Local-first
Aggregate-only sync

100%

Screen 2 — Laboratories. Site list with the active context marked. Selecting a different laboratory re-scopes every other workspace immediately.

4.1 Fields that matter for national reporting

FIELD	NOTES
Code, name	The code is the laboratory's identity in exports and federation. It appears as an <code>Organization.identifier</code> in FHIR and in MSH-4 of every HL7 message.
Country	Chosen from the coded country list — ISO 3166-1 alpha-3 with alpha-2 and WHO region held as metadata.
State / UT, District	Selected by name but stored as LGD codes — the Local Government Directory identifiers maintained by the Ministry of Panchayati Raj. 36 states and union territories and 785 districts ship in the catalogue. The district list filters to the chosen state.
Site group	Free text used to group sites into a network (for example an AMRSN cohort). Exported as a second <code>Organization.identifier</code> .
Interpretation defaults	Default guideline and test method, dynamic-breakpoint and intrinsic-resistance switches, enabled expert rules, half-dilution rounding, conditional reporting and clinical-message printing.

Edit laboratory

Required identity and default AST settings. Location codes can also be managed in Master Studio.

Stable identifier used in records and federation.

Laboratory code *
HARP

Laboratory name *
Harpreet

Country
India

State / UT
Punjab

District
Amritsar

State LGD code
3

District LGD code
27

One Health domain
Custom...

Default guideline
CLSI

One Health domain — custom value
ICMR Punjab validation

Default test method
Disk diffusion

Guideline year / edition
2025

Cancel Save laboratory

Screen 3 — Laboratory editor. State and district are chosen by name; the LGD code is written alongside and is also editable directly, so a site can be reconciled against a national directory without re-typing the name.

Why LGD codes matter

Indian health programmes reconcile facility data on LGD codes rather than place names, because district names change and repeat across states. AMRIT stores the code and the name together, so an export carries an unambiguous geography even when the display name is later revised. The LGD code set is the identifier family referenced by the Metadata and Data Standards (MDDS) demographic domain, which makes it the correct binding for Indian national reporting.

4.2 Cloning a laboratory

A laboratory can be duplicated with its configuration — antibiotic settings, panels, alerts and enabled rules — but without its isolates. This is the supported way to bring a second site online with an identical validated configuration.

5 · Isolate records

The isolate editor is the primary capture surface. It is code-first: choosing an organism or specimen selects a master entry, and that selection drives panel matching, breakpoint interpretation and every downstream export.

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ACTIVE LABORATORY
Harpreet · HARP

SYNC OFFLINE Settings

SURVEILLANCE WORKFLOW

Isolate records

Enter, validate and review antimicrobial susceptibility records with master-driven panels and duplicate safeguards.

+ New isolate Help

11000 FINAL 0 DRAFT 11000 SHOWN

Search specimen, patient or organism... All statuses Refresh

SPECIMEN	PATIENT ID	ORGANISM	LOCATION	AST	STATUS	ACTIONS
SYN-HARP-009992 2026-07-10 · Sputum	SYN-009992	Klebsiella pneumoniae 56415008	Emergency Room out	9 results	FINAL	Review Export
SYN-HARP-009856 2026-07-10 · Urine	SYN-009856	Escherichia coli 112283007	Surgical Ward in	9 results	FINAL	Review Export
SYN-HARP-009843 2026-07-10 · Blood	SYN-009843	Staphylococcus aureus 3052008	Emergency Room out	5 results	FINAL	Review Export
SYN-HARP-009652 2026-07-10 · Urine	SYN-009652	Pseudomonas aeruginosa 52499004	Surgical Ward in	6 results	FINAL	Review Export
SYN-HARP-008951 2026-07-10 · Urine	SYN-008951	Pseudomonas aeruginosa 52499004	Intensive Care Unit icu	6 results	FINAL	Review Export
SYN-HARP-008334 2026-07-10 · Blood	SYN-008334	Escherichia coli 112283007	Emergency Room out	9 results	FINAL	Review Export
SYN-HARP-008174 2026-07-10 · Catheter	SYN-008174	Escherichia coli 112283007	Intensive Care Unit icu	9 results	FINAL	Review Export
SYN-HARP-007786 2026-07-10 · Urine	SYN-007786	Enterococcus faecium 90272000	Medical Ward in	3 results	FINAL	Review Export
SYN-HARP-006509 2026-07-10 · Urine	SYN-006509	Klebsiella pneumoniae 56415008	Medical Ward in	9 results	FINAL	Review Export
SYN-HARP-005124 2026-07-10 · Sputum	SYN-005124	Enterococcus faecium 90272000	Surgical Ward in	3 results	FINAL	Review Export
SYN-HARP-004743 2026-07-10 · Tissue	SYN-004743	Acinetobacter baumannii 91288006	Surgical Ward in	6 results	FINAL	Review Export
SYN-HARP-001224 2026-07-10 · Sputum	SYN-001224	Pseudomonas aeruginosa 52499004	Intensive Care Unit icu	6 results	FINAL	Review Export

Screen 4 — Isolate records. 11 000 records in the active laboratory, all final. The organism column shows the display name over the stored code — here SNOMED CT concept identifiers such as **56415008** and **112283007**. Filters cover free-text search and record status.

5.1 Creating and completing a record

1. **Identity and encounter** — patient identifier, specimen number, specimen date, sex, date of birth, location, location type, department, admission date, diagnosis code.
2. **Specimen and organism** — chosen from the master catalogues. Selecting from the list writes both the code and the display name, which is what keeps the pair internally consistent.
3. **Panel** — as soon as organism, specimen and location type are known, AMRIT matches active AST panels and preselects the best fit. The panel prescribes which antibiotics appear.
4. **AST results** — one row per antibiotic with measurement, interpretation, method and guideline.
5. **Genotypic markers and omics** — optional molecular determinants and identification metadata.
6. **Save as draft or final** — the final gate is enforced in the main process, not only in the form.

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ACTIVE LABORATORY

Harpreeet · HARP

SYNC OFFLINE

Settings

AMC - added for this record

27

S — Susceptible

Disk diffusion (CLSI)

CLSI

Amikacin

AMK - added for this record

30

S — Susceptible

Custom...

Disk diffusion (CLSI)

CLSI

Ceftriaxone

CRO - added for this record

34

S — Susceptible

Custom...

Disk diffusion (CLSI)

CLSI

Ceftazidime

CAZ - added for this record

8

R — Resistant

Custom...

Disk diffusion (CLSI)

CLSI

Ciprofloxacin

CIP - added for this record

10

R — Resistant

Custom...

Disk diffusion (CLSI)

CLSI

Colistin

COL - added for this record

22

S — Susceptible

Custom...

Disk diffusion (CLSI)

CLSI

Gentamicin

GEN - added for this record

35

S — Susceptible

Custom...

Disk diffusion (CLSI)

CLSI

Meropenem

MEN - added for this record

34

S — Susceptible

Custom...

Disk diffusion (CLSI)

CLSI

Trimethoprim/Sulfamethoxazole

SXT - added for this record

12

R — Resistant

Custom...

Disk diffusion (CLSI)

CLSI

Documented no-AST reason

Not applicable — AST entered

Screen 5 — Isolate editor, upper half. Identity, encounter and specimen fields. The organism and specimen selectors write a code and a display name together; the panel selector below them re-matches as soon as that pair changes.

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Harpreeet · HARP

SYNC OFFLINE

Settings

Applied to every row in this record and to antibiotics added below.

Stored with each result as interpretation provenance.

Test method — custom value

Disk diffusion (CLSI)

ANTIBIOTIC	MEASUREMENT	INTERPRETATION	METHOD	GUIDELINE
Amoxicillin/Clavulanic acid AMC - added for this record	27	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI
Amikacin AMK - added for this record	30	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI
Ceftriaxone CRO - added for this record	34	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI
Ceftazidime CAZ - added for this record	8	R — Resistant	Custom... Disk diffusion (CLSI)	CLSI
Ciprofloxacin CIP - added for this record	10	R — Resistant	Custom... Disk diffusion (CLSI)	CLSI
Colistin COL - added for this record	22	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI
Gentamicin GEN - added for this record	35	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI
Meropenem MEN - added for this record	34	S — Susceptible	Custom... Disk diffusion (CLSI)	CLSI

Screen 6 — AST entry. Each antibiotic keeps its measurement, its S/I/R interpretation, its test method and its guideline. The provenance is stored with the result, which is why an exported interpretation can always be traced to the rule that produced it.

The measurement column has no unit

A disk-diffusion zone of 27 mm and an MIC of 27 mg/L are both entered as 27 ; only the adjacent method field distinguishes them. Inside AMRIT this is unambiguous because the method travels with the value. It becomes a problem at the export boundary — see Finding F-03 in Chapter 22.

The measurement column has no unit

A disk-diffusion zone of 27 mm and an MIC of 27 mg/L are both entered as 27 ; only the adjacent method field distinguishes them. Inside AMRIT this is unambiguous because the method travels with the value. It becomes a problem at the export boundary — see Finding F-03 in Chapter 22.

5.2 The final-record gate

A record may only be marked final when it has a patient identifier, a specimen date, a specimen type or code, and an organism name or code. Anything short of that is saved as a draft. Drafts are excluded from analytics denominators and from every federated answer.

5.3 Duplicate detection

Before insert, AMRIT looks for an existing record with the same patient identifier, specimen number, specimen date and organism. A match is refused rather than merged, so a re-import cannot silently double a denominator.

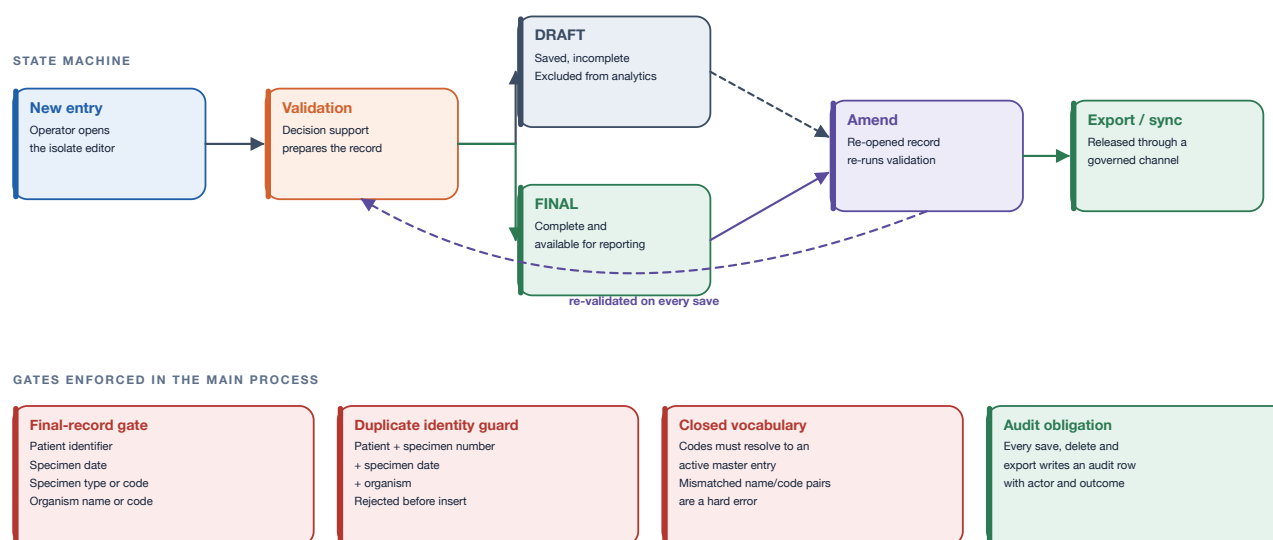
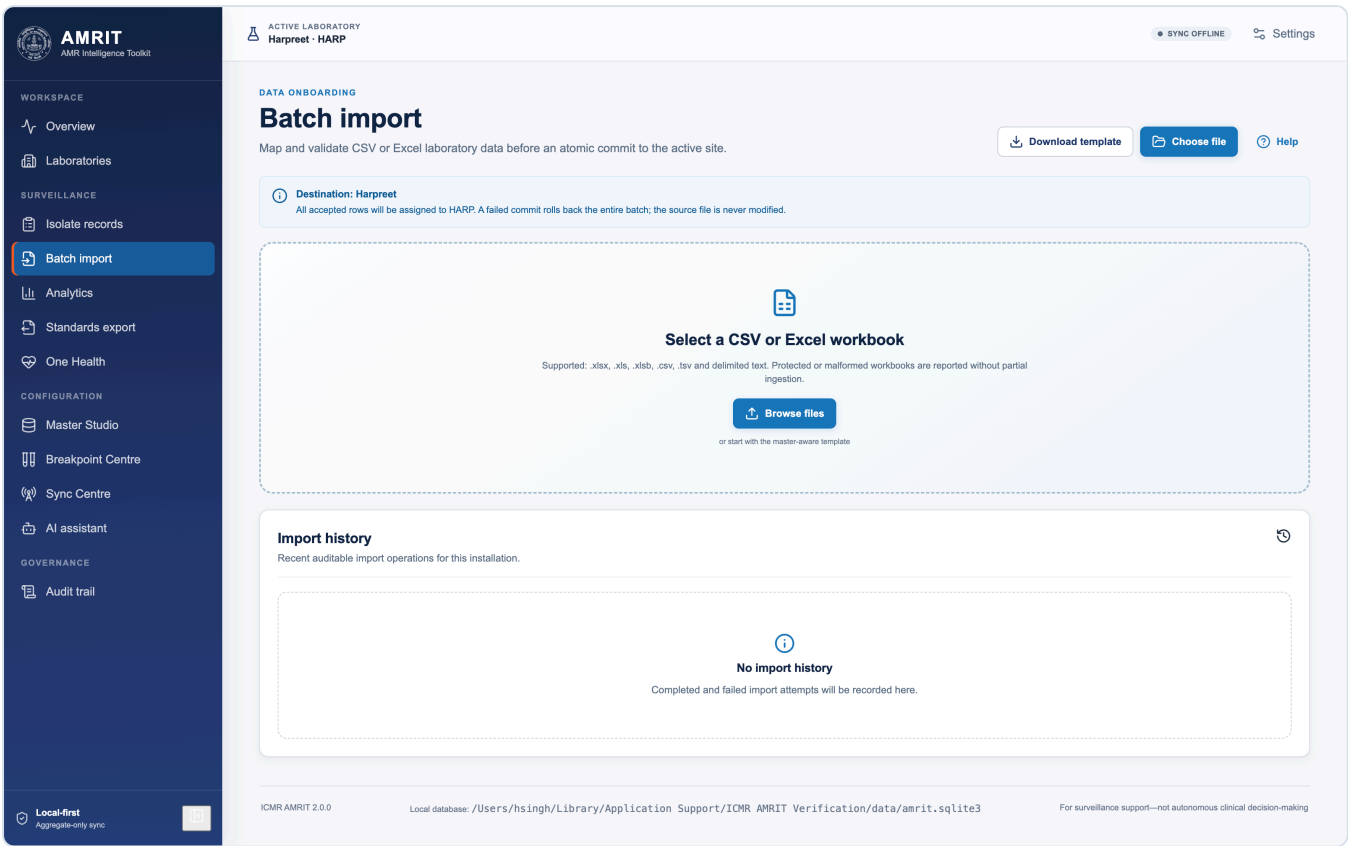


Figure 2 — Isolate record lifecycle. States, transitions and the four gates the main process enforces regardless of what the interface sends.

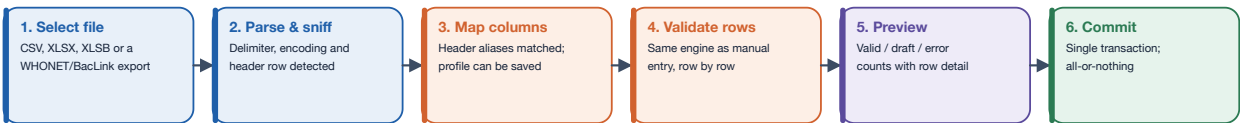
6 · Batch import

Import runs the same validation engine as manual entry, previews the entire outcome before anything is written, and then commits in a single transaction.



Screen 7 — Batch import. Source selection, mapping and preview in one workspace. Mappings can be saved as a named profile and reused for a recurring feed.

SIX STEPS, ONE TRANSACTION



HEADER ALIASING

patient id / patient / mrn	→	patient_id
specimen no / accession / lab no	→	specimen_number
collection date / spec date	→	specimen_date
organism / isolate / pathogen	→	organism
organism code / organism id	→	organism_code
diagnosis code / icd / icd code	→	diagnosis_code
Unmapped columns are preserved as custom fields.		

WHAT THE PREVIEW REPORTS

Valid Rows that will be committed as final records	Draft Rows missing a final-record requirement
Error Rows blocked by the closed vocabulary or a contradictory name/code pair. Each issue carries the row number, field and message so the source file can be corrected and re-previewed before any data is written.	

Import-time caution for national reporting

Date parsing assumes DD/MM/YYYY when a slash-separated value is ambiguous — confirm the source convention before import.
Organism and specimen codes are written to a single code column. If the source file supplies SNOMED CT identifiers they will be stored as the working code; if it supplies WHONET mnemonics those are stored instead. Standardise the source file so one laboratory does not mix code systems across batches — see Chapter 9 for the interoperability consequence.

Figure 3 — Batch import pipeline. Six steps, one transaction, and the header aliases the mapper recognises automatically.

6.1 Supported sources

- Delimited text (CSV and variants) with delimiter and encoding detection.
- Excel workbooks in XLSX and XLSB.
- WHONET and BacLink style exports, including WHONET laboratory configuration files, which can populate antibiotic settings, locations, data fields and profiles rather than isolates.

6.2 Reading the preview

BUCKET	MEANING	WHAT TO DO
Valid	Row satisfies the final-record gate and every code resolves.	Nothing — it will be committed as final.
Draft	Row is well-formed but incomplete.	Commit and complete in the editor, or fix the source and re-preview.
Error	An unknown or inactive code, or a name and code that resolve to different master entries.	Correct the source file. Errors block the row, not the batch.

Two import-time cautions for national reporting

- **Ambiguous dates.** A slash-separated date is read as DD/MM/YYYY. `03/04/2026` is therefore 3 April, not 4 March. Confirm the source convention before importing.
- **Code-system drift.** Organism and specimen codes are written to a single column. If one feed supplies SNOMED CT concept identifiers and another supplies WHONET mnemonics, the same laboratory will hold both in the same field — visible in the live database used for this verification. Standardise the source before import; see Finding F-01.

7 · Decision support and interpretation

Every save — manual or imported — passes through the same six-stage pipeline in the main process. The interface cannot bypass it.

EVERY SAVE — MANUAL OR IMPORTED — TRAVERSES THIS PIPELINE

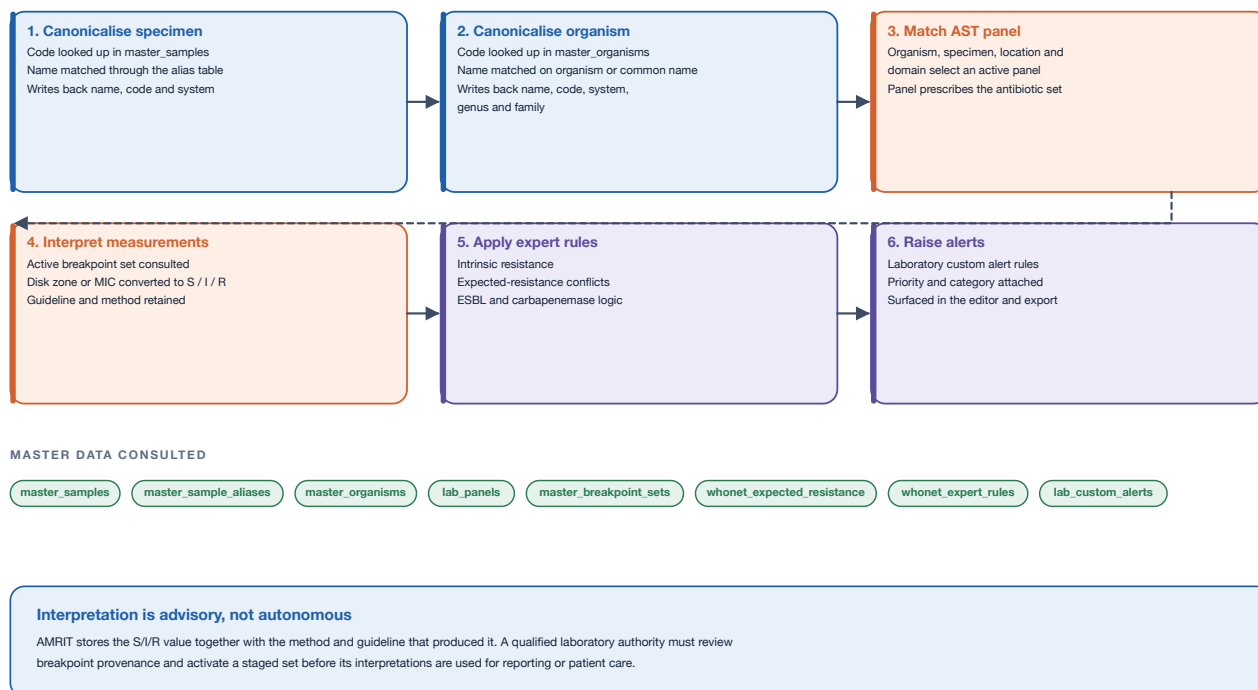


Figure 4 — Decision-support pipeline. Canonicalisation, panel matching, interpretation, expert rules and alerting, with the master tables each stage consults.

7.1 Canonicalisation

Specimen and organism are resolved from whatever the record supplies. A code is looked up directly; a name is matched against the master name, the common name and — for specimens — an alias table that carries 38 spelling and phrasing variants. Three outcomes are possible:

- **Resolved.** The record is rewritten with the canonical name, code and code system, plus genus and family for organisms.
- **Unknown.** The code or name does not match an active master entry. This is an error on final records.
- **Contradictory.** The supplied name and code resolve to *different* active masters. Always a hard error — AMRIT will not guess which one the operator meant.

7.2 Panel matching and interpretation

The canonical organism, specimen, location type and domain select an active AST panel, which prescribes the antibiotic set. Measurements are then interpreted against the *activated* breakpoint set for the guideline and method in force, and the guideline and method are stored alongside the resulting S/I/R value.

7.3 Expert rules and alerts

Seven packaged expert rules ship with the catalogue, together with 1 024 expected-resistance entries. Rules cover intrinsic resistance, ESBL and carbapenemase logic, and expected-resistance conflicts. Laboratory-

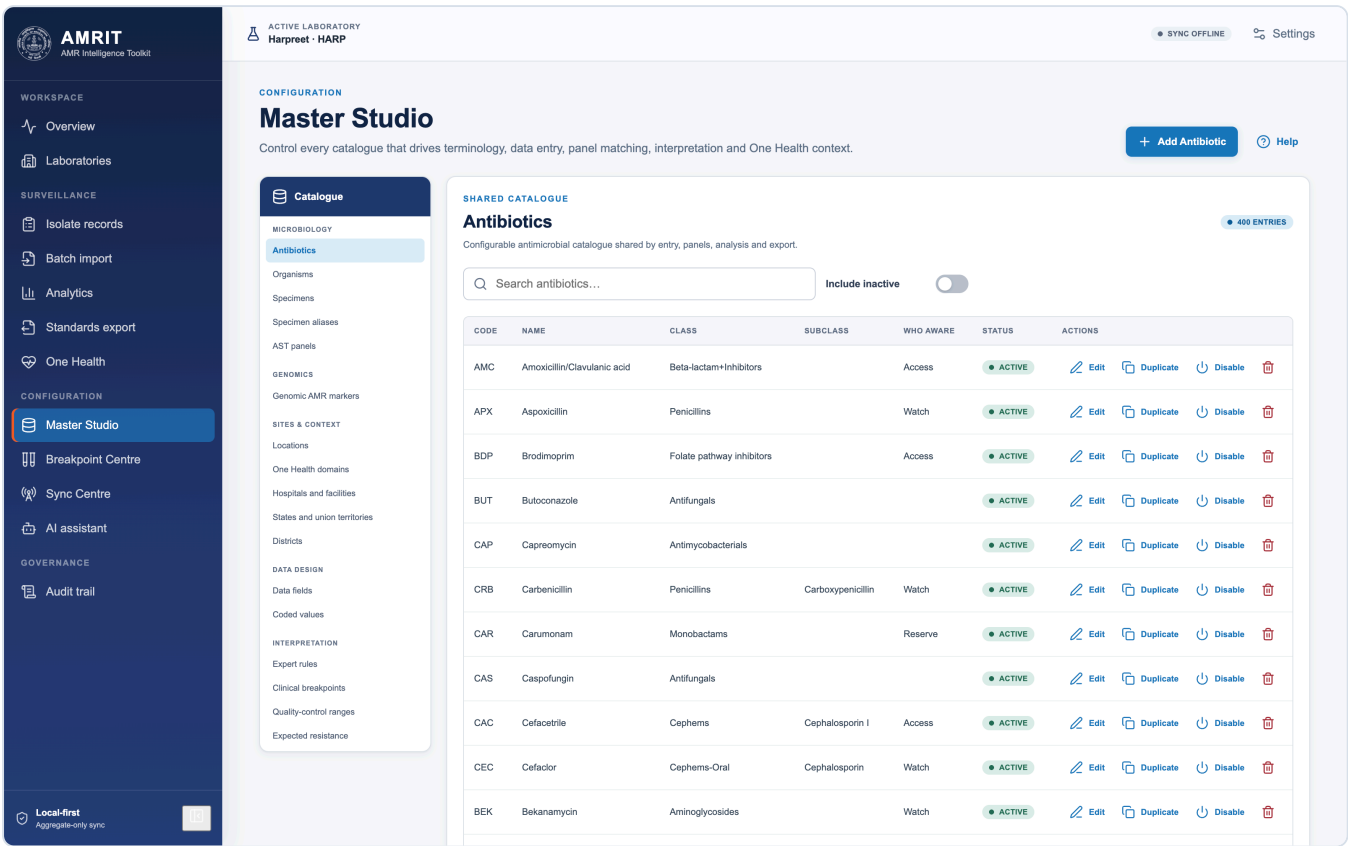
specific alert rules add local triggers with their own category, priority and message. Rule output is advisory: it annotates the record and appears in exports, but it never overwrites an operator's interpretation.

Authority remains with the laboratory

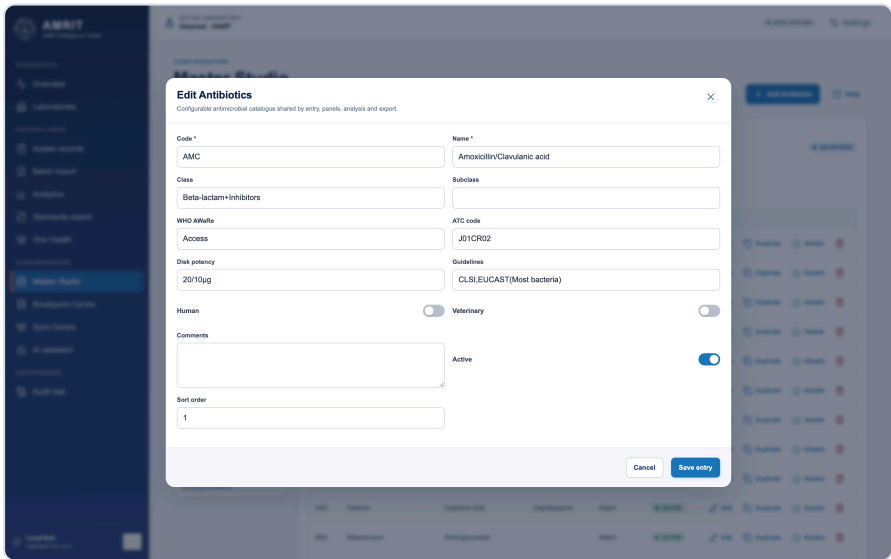
AMRIT stores an S/I/R value together with the method and guideline that produced it. It does not authorise a result. A staged breakpoint set never interprets anything until a laboratory authority activates it deliberately.

8 • Master Studio

Seventeen catalogues, one editing model. Master Studio is where the application's terminology actually lives, and where a site adapts national content to local practice without forking the software.



Screen 8 — Master Studio, antibiotics. 400 entries with class, subclass and WHO AWaRe group. Search covers code, name, class, AWaRe group and ATC code.



Screen 9 — Antibiotic editor. The WHO ATC code, the AWaRe group and both LOINC susceptibility codes are editable fields on every entry. The application holds this terminology; the finding recorded in Chapter 22 is that it does not yet *emit* it.

8.1 The catalogues

CATALOGUE	ROWS SHIPPED	TERMINOLOGY CARRIED
Antibiotics	399	WHONET code, WHO ATC, LOINC MIC and disk codes, WHO AWaRe, EUCAST/DIN/JAC codes, class and subclass
Organisms	2 380	WHONET code, SNOMED CT concept, GBIF taxon, full Linnaean rank chain, taxonomic status
Specimens	8	Local code, SNOMED CT concept where available, parent code for hierarchy
Specimen aliases	38	Normalised alias text mapped to a specimen code
Coded values	2 710	Twelve code sets — see §8.3
States and union territories	36	LGD code, union-territory flag, country
Districts	785	LGD code linked to the state LGD code
Panels	43	Organism, specimen and antibiotic membership with requirement type and choice logic
Expected resistance	1 024	Organism/antibiotic pairs with code types and guideline attribution
Expert rules	7	Rule code, criteria, affected antibiotics and exceptions
MIC panels	77	Named dilution ranges per antibiotic
Field definitions	898	WHONET field metadata across human, animal, food, feed, plant and environment modules
Genomic markers	52	Gene symbol, mechanism class, predicted resistance, linked antibiotics, reporting standard
Also editable: locations, domains, laboratory data fields, hospitals, breakpoints and QC ranges.		

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Master Studio

Control every catalogue that drives terminology, data entry, panel matching, interpretation and One Health context.

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SHARED CATALOGUE

Organisms

Taxonomy-aware organism catalogue used for code-first panel matching and surveillance.

• 2390 ENTRIES

Q Search organisms...

Include inactive ☐

CODE	ORGANISM	COMMON NAME	TYPE	SNOMED CT	STATUS	ACTIONS
SAJ	Abiotrophia adiacens		+	113713009	ACTIVE	Edit Duplicate Disable Delete
SDF	Abiotrophia defectiva		+	113714003	ACTIVE	Edit Duplicate Disable Delete
KPN	Klebsiella pneumoniae complex				ACTIVE	Edit Duplicate Disable Delete
ABI	Abiotrophia sp.		+	372391001	ACTIVE	Edit Duplicate Disable Delete
ACF	Abisidia corimbifera		f		ACTIVE	Edit Duplicate Disable Delete
ABS	Abisidia sp.		f	404226006	ACTIVE	Edit Duplicate Disable Delete
EAE	Klebsiella aerogenes				ACTIVE	Edit Duplicate Disable Delete
ACN	Acanthamoeba sp.		w	372401005	ACTIVE	Edit Duplicate Disable Delete
AVB	Acetivibrio sp.		a	131199002	ACTIVE	Edit Duplicate Disable Delete
ACE	Acetobacterium sp.		a	423008000	ACTIVE	Edit Duplicate Disable Delete
ACP	Acholeplasma sp.		b	131200004	ACTIVE	Edit Duplicate Disable Delete

Local-first
Aggregate-only sync

Screen 10 — Organism catalogue. 2 390 entries in this installation. The SNOMED CT column is a first-class field, not an afterthought — and the gaps in it are visible, which is what makes coverage auditable.

Edit Organisms

Taxonomy-aware organism catalogue used for code-first panel matching and surveillance.

Code * SAJ

Organism * Abiotrophia adiacens

Common name

Type +

SNOMED CT 113713009

GBIF taxon 3227166

Kingdom Bacteria

Family Carnobacteriaceae

Genus Granulicatella

Anaerobe ☐

Common commensal ☐ Active ☒

Sort order 1

[Cancel](#) [Save entry](#)

Screen 11 — Organism editor. SNOMED CT concept identifier and GBIF taxon identifier alongside kingdom, family and genus. A site can correct or supply a standard code without a software change.

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SHARED CATALOGUE

Specimens

Standard specimen catalogue with hierarchy and terminology codes.

Q

Search specimens...

Include inactive

8 ENTRIES

CODE	NAME	PARENT CODE	CODE SYSTEM	CONCEPT CODE	STATUS	ACTIONS
URINE	Urine	—	http://snomed.info/sct	122575003	ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
BLOOD_STERILE	Blood / normally sterile fluid	—			ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
CSF	Cerebrospinal fluid	—	http://snomed.info/sct	258450006	ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
LOWER_RESP	Lower respiratory tract	—			ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
DEEP_TISSUE_WOUND	Deep tissue / wound / musculoskeletal	—			ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
INTRA ABDOMINAL DEEP	Intra-abdominal / deep sterile specimen	—			ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
ENTERIC_STOOL	Stool / enteric	—	http://snomed.info/sct	119339001	ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove
SEXUAL_HEALTH_ISOLATE	Genital / extragenital sexual-health isolate	—			ACTIVE	<a>Edit <a>Duplicate <a>Disable <a>Remove

<

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>

Screen 12 — Specimen catalogue. Eight entries, of which three carry a SNOMED CT concept identifier. The remaining five are grouping concepts — *blood or normally sterile fluid*, *lower respiratory tract*, *deep tissue / wound / musculoskeletal*, *intra-abdominal* and *genital / extragenital sexual-health isolate* — and there is no single SNOMED concept that means exactly what each of them means.

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Genomic AMR markers

Genotypic resistance determinants, molecular assays and typing outputs available to AST panels and isolate entry.

Q Search genomic amr markers...

Include inactive

CODE	MARKER	MARKER TYPE	GENE / LOCUS	MECHANISM	STATUS	ACTIONS
BLANDM	blaNDM (New Delhi metallo-β-lactamase)	gene	blaNDM	Carbapenemase (Ambler class B, MBL)	ACTIVE	Edit Duplicate Disable Delete
BLAKPC	blaKPC (Klebsiella pneumoniae carbapenemase)	gene	blaKPC	Carbapenemase (Ambler class A)	ACTIVE	Edit Duplicate Disable Delete
BLAOXA48	blaOXA-48-like (incl. OXA-181, OXA-232)	gene	blaOXA-48-like	Carbapenemase (Ambler class D)	ACTIVE	Edit Duplicate Disable Delete
BLAVIM	blaVIM (Verona integron-encoded MBL)	gene	blaVIM	Carbapenemase (Ambler class B, MBL)	ACTIVE	Edit Duplicate Disable Delete
BLAIMP	blaIMP (imipenemase MBL)	gene	blaIMP	Carbapenemase (Ambler class B, MBL)	ACTIVE	Edit Duplicate Disable Delete
BLAOXA23	blaOXA-23-like	gene	blaOXA-23-like	Carbapenem-hydrolysing class D β-lactamase	ACTIVE	Edit Duplicate Disable Delete
BLAOXA24	blaOXA-24/40-like	gene	blaOXA-24/40-like	Carbapenem-hydrolysing class D β-lactamase	ACTIVE	Edit Duplicate Disable Delete
BLAOXA58	blaOXA-58-like	gene	blaOXA-58-like	Carbapenem-hydrolysing class D β-lactamase	ACTIVE	Edit Duplicate Disable Delete
BLAOXA51	blaOXA-51-like (species marker)	identification	blaOXA-51-like	Intrinsic class D β-lactamase used for species confirmation	ACTIVE	Edit Duplicate Disable Delete
BLACTXM	blaCTX-M (group 1 / group 9)	gene	blaCTX-M	Extended-spectrum β-lactamase (ESBL)	ACTIVE	Edit Duplicate Disable Delete

Screen 13 — Genomic AMR markers. 52 determinants with gene symbol, mechanism class, predicted resistance, linked antibiotic codes and the reporting standard each is drawn from — WHO GLASS-WGS, the WHO Bacterial Priority Pathogens List and ICMR AMRSN.

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States and union territories

LGD-aligned state master for laboratory and facility geography.

Q Search states and union territories...

Include inactive

LGD CODE	STATE/UT	COUNTRY	UNION TERRITORY	ACTIVE	STATUS	ACTIONS
35	Andaman And Nicobar Islands	India	YES	YES	ACTIVE	Edit Duplicate Disable Delete
28	Andhra Pradesh	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
12	Arunachal Pradesh	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
18	Assam	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
10	Bihar	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
4	Chandigarh	India	YES	YES	ACTIVE	Edit Duplicate Disable Delete
22	Chhattisgarh	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
38	The Dadra And Nagar Haveli And Daman And Diu	India	YES	YES	ACTIVE	Edit Duplicate Disable Delete
7	Delhi	India	YES	YES	ACTIVE	Edit Duplicate Disable Delete
30	Goa	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete
24	Gujarat	India	NO	YES	ACTIVE	Edit Duplicate Disable Delete

Screen 14 — States and union territories. 36 rows keyed on the LGD code, with the union-territory flag. Districts are a second catalogue of 785 rows keyed on their own LGD code and linked to the state code.

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8.2 Protected catalogues

Antibiotics, organisms, specimens and aliases are marked as protected. Packaged rows carry their source dataset and version, and a `user_modified` flag records local divergence — so a future catalogue refresh can distinguish deliberate local edits from stale content.

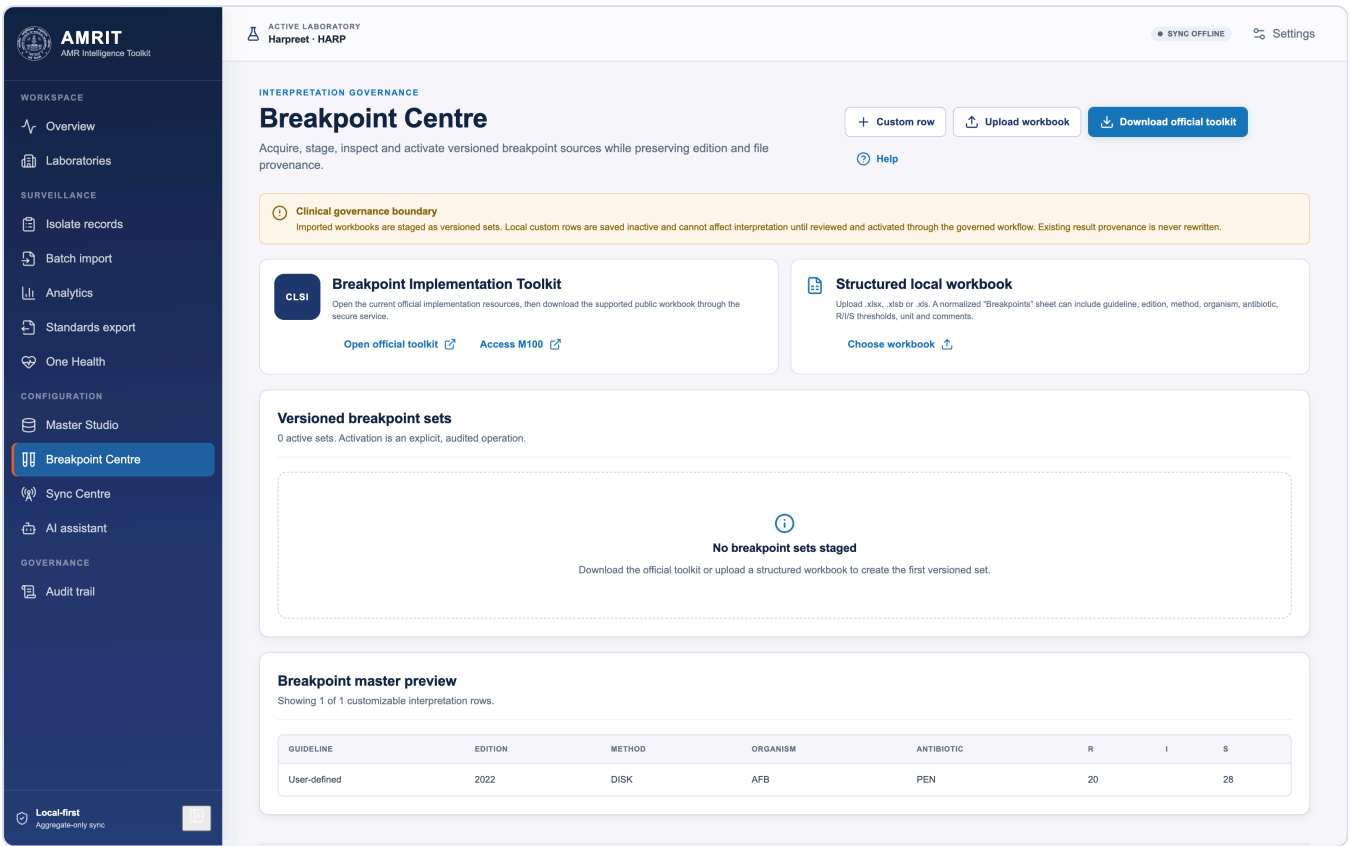
8.3 Coded value sets

CODE SET	ROWS	CODE SET	ROWS	CODE SET	ROWS
serotype	2 150	country	252	food_category	84
species	76	ward_type	43	department	29
animal_type	18	species_group	18	specimen_reason	17
sex_category	9	patient_type	8	market_category	6

Country rows carry ISO 3166-1 alpha-3 as the code, with alpha-2 and the WHO region in the metadata payload — for example `AFG` with `{"iso2":"AF","who_region":"EMRO"}`.

9 · Breakpoint Centre

Breakpoints are the most consequential configuration in the application, so they are the most tightly governed: staged, hashed, reviewed and activated as a deliberate act.



Screen 15 — Breakpoint Centre. Versioned sets with guideline, year, method and scope; a custom-rule form for site-specific entries; and links to the official CLSI and EUCAST sources.

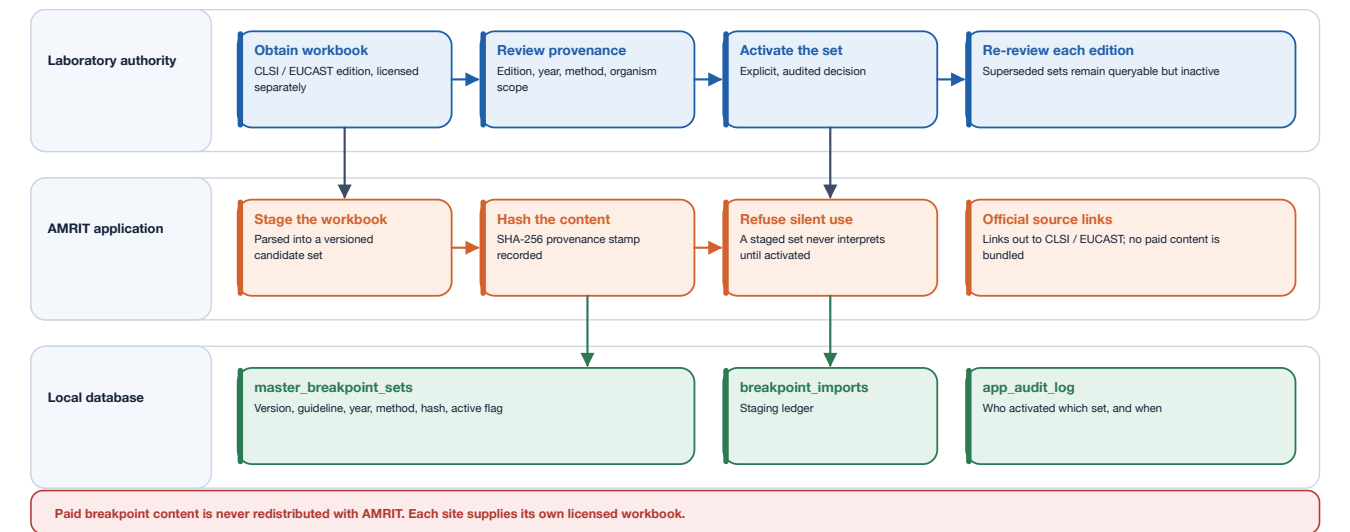


Figure 5 — Breakpoint governance lifecycle. Three lanes: what the laboratory authority does, what the application does, and what is written to the database.

9.1 The workbook contract

1. The laboratory obtains a breakpoint workbook for the edition it is licensed to use.
2. AMRIT parses it into a versioned candidate set and records a SHA-256 provenance hash of the content.
3. The staged set is inert. It interprets nothing.
4. A reviewer inspects edition, year, method, organism scope and hash, then activates the set explicitly. The activation is audited with the actor.
5. Superseded sets remain queryable but inactive, so a historical interpretation can still be explained.

Licensing

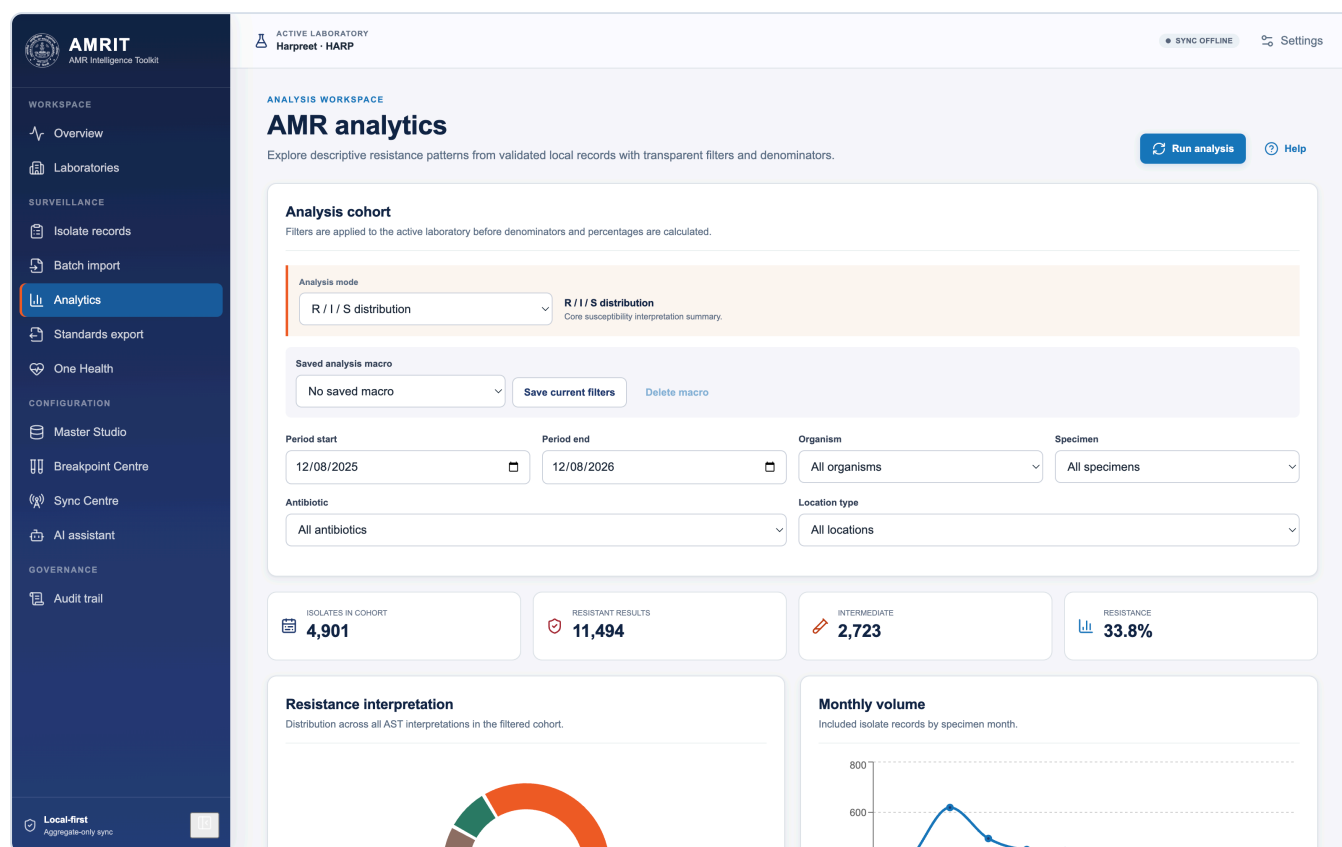
Paid breakpoint content is not bundled with AMRIT and is not redistributed by it. The application links to the official CLSI and EUCAST sources; each site supplies its own licensed workbook.

9.2 Custom rules

A site can add individual breakpoint rows — guideline, year, test method, breakpoint type, organism code, antibiotic code, R/I/S values, potency and comments — for organisms or agents its workbook does not cover. Custom rows are inactive until enabled, and are flagged as custom everywhere they appear.

10 · Analytics

Ten deterministic analysis modes over the filtered local cohort. No sampling, no model, no hidden smoothing — every number is recomputed from the records the filters select.



Screen 16 — Analytics. A real cohort: 4 901 isolates, 11 494 resistant results, 2 723 intermediate, 33.8 % resistance. Filters, denominators and the resulting figures are shown together so a number can never be quoted without its cohort.

10.1 The modes

MODE	PURPOSE
R / I / S distribution	Core susceptibility interpretation summary with per-antibiotic denominators.
Antimicrobial stewardship	AWaRe-oriented view of the agents in use.
Unit antibiogram	Susceptibility by ward or location type.
Specimen antibiogram	Susceptibility by specimen category.
Priority indicators	Programme-priority organism/agent combinations.
Data quality & contamination	Missingness, contaminant flags and review counts.
Temporal trends	Monthly volume and resistance movement.
Isolate listing	Row-level listing of the filtered cohort.
Organism / specimen summary	Composition of the cohort.
Cluster watch	Candidate clusters for review.

10.2 Denominator rules

- Draft records are excluded unless drafts are explicitly included.
- Resistance percentage uses S, I and R as the denominator and R as the numerator.
- Deduplication can be set to first isolate per patient and organism, or all isolates.
- Small denominators are shown, not hidden, with the tested count beside every rate.

Filter sets can be saved as named analysis macros and reused, which is how a site keeps a monthly report reproducible.

11 · Standards export

Six formats, one cohort definition. The export cohort is the active laboratory filtered by an optional date range, and the cohort is recorded in the audit entry.

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INTEROPERABILITY
Standards export
Create traceable WHONET, FHIR, HL7 and analysis files from a clearly filtered local cohort. [Help](#)

Export cohort
Leave dates blank for all available records at the active laboratory.

Period start: dd/mm/yyyy Period end: dd/mm/yyyy

WHONET
WHONET-compatible isolate/AST table for established AMR workflows.
- .csv
Save export

FHIR R4 Bundle
Standards-coded diagnostic report and observation resources.
- .json
Save export

HL7 v2.5.1
ORU-style pipe-delimited laboratory messages with OBX interpretation.
- .hl7
Save export

FHIR MeasureReport
Aggregate surveillance measures suitable for federated reporting.
- .json
Save export

Analysis CSV
Flat, analysis-ready export of the filtered local cohort.
- .csv
Save export

AMRIT JSON
Loss-minimising structured backup/interchange representation.
- .json
Save export

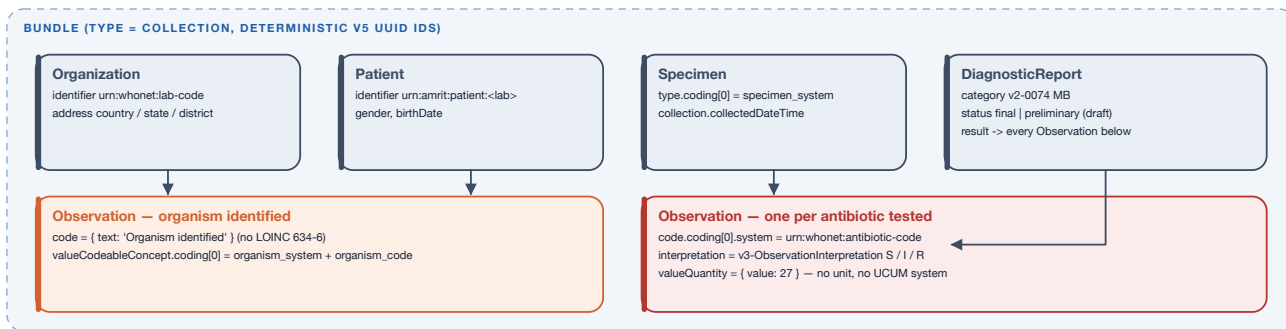
Interoperability safeguards
What the export pipeline preserves.

- Coded identities**
Organism, specimen and antibiotic codes remain attached where the target format supports them.
- Format-specific AST**
Structured exports retain detailed AST fields; WHONET CSV emits result and measurement columns.
- Explicit cohort**
Laboratory and date filters are audited with the export.

Screen 17 — Standards export. Each format states what it preserves. The interoperability-safeguards panel at the foot of the page sets the expectation that coded identities survive “where the target format supports them”.

FORMAT	FILE	CONTENT
WHONET	• .csv	Isolate and AST table with a result and a measurement column per antibiotic, plus method, guideline, potency and source cohort.
FHIR R4 Bundle	• .json	Organization, Patient, Specimen, Observation (organism identification and one per antibiotic) and DiagnosticReport, as a collection bundle with UUID URNs.
HL7 v2.5.1	• .hl7	ORU-style messages: MSH, PID, PV1, SPM, OBR, one OBX for the organism, one OBX per antibiotic, and NTE notes.
FHIR MeasureReport	• .json	A proportion measure for one antibiotic with initial-population, denominator and numerator counts, a percentage in UCUM, and an organism stratifier.
Analysis CSV	• .csv	Flat analysis-ready extract of the filtered cohort.
AMRIT JSON	• .json	Loss-minimising structured representation for backup and interchange.

ONE ISOLATE RECORD, AS IT LEAVES THE DEVICE IN A FHIR R4 BUNDLE



ELEMENT-BY-ELEMENT VERDICT AGAINST FHIR R4 AND ITS BOUND CODE SYSTEMS

● Bundle.entry.fullUrl / resource.id	urn:uuid + RFC 4122 v5, deterministic and stable
● Observation.interpretation	http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation
● Observation.category	http://terminology.hl7.org/CodeSystem/observation-category = laboratory
● DiagnosticReport.category	http://terminology.hl7.org/CodeSystem/v2-0074 = MB
● Specimen.type.coding.system	SNOMED CT when the record carries a concept id; a local URI otherwise
● Observation.valueCodeableConcept	SNOMED CT declared, but the code column may hold a WHONET mnemonic
● Observation.code (susceptibility)	urn:whonet:antibiotic-code; the LOINC codes in the catalogue are not used
● Observation.valueQuantity	no unit / system / code — mm and mg/L are indistinguishable to a receiver
● Resource.meta.profile	absent — the Bundle claims conformance to no implementation guide
● Patient.identifier.system	urn:amrit:patient:<lab> — not an ABHA / ABDM namespace

Green — conformant and bound to a published code system. Amber — conformant only for part of the data. Red — a receiving system cannot safely resolve the value.

Figure 6 — Anatomy of an exported FHIR R4 Bundle. The five resource types one isolate produces, how they reference one another, and an element-by-element verdict against FHIR R4 and the code systems it binds. The verdicts are reproduced from a Bundle produced by the compiled build against the reference database; Chapter 22 gives the evidence for each red row.

11.1 What is guaranteed

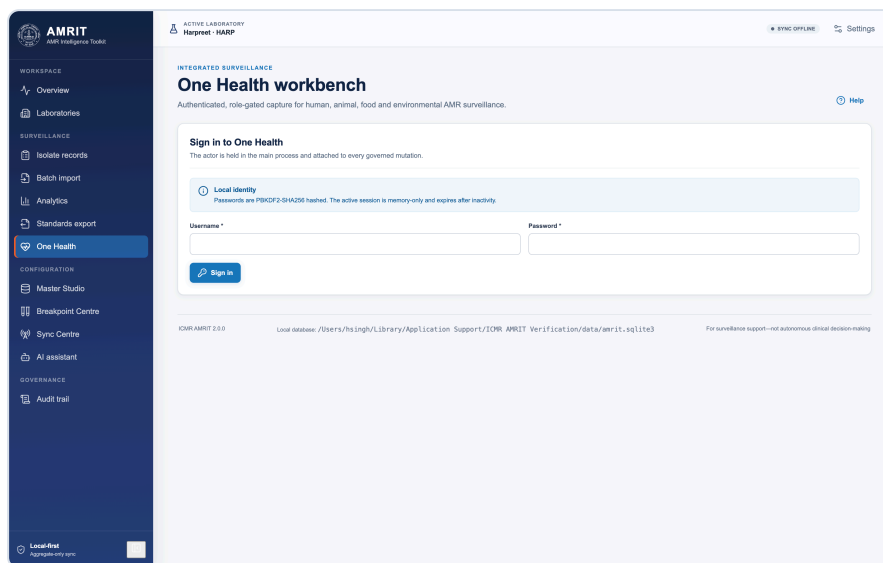
- The destination is a path the operator chose in a save dialog; the renderer cannot supply one.
- Every export writes an audit row with the format, the record count, the laboratory and the file name.
- FHIR resource identifiers are deterministic RFC 4122 version-5 UUIDs, so re-exporting the same cohort produces the same identifiers. The bundle builder asserts this and refuses to emit a resource whose identifier is not a v5 UUID.

What is *not* guaranteed

The exports are structurally valid but terminologically incomplete: no LOINC code is attached to any observation, no UCUM unit is attached to any measurement, and the WHO ATC codes and AWaRe groups held in the catalogue are not carried out. Chapter 21 states the position for each standard and Chapter 22 gives the evidence.

12 · One Health workbench

A separately authenticated, role-gated workspace for cross-sector surveillance. It has its own identities, its own permission model and its own tamper-evident audit log.



Screen 18 — One Health sign-in. Passwords are PBKDF2-SHA256 hashed. The session is held in the main process, is memory-only, and expires after 30 minutes of inactivity.

12.1 First run: creating the administrator

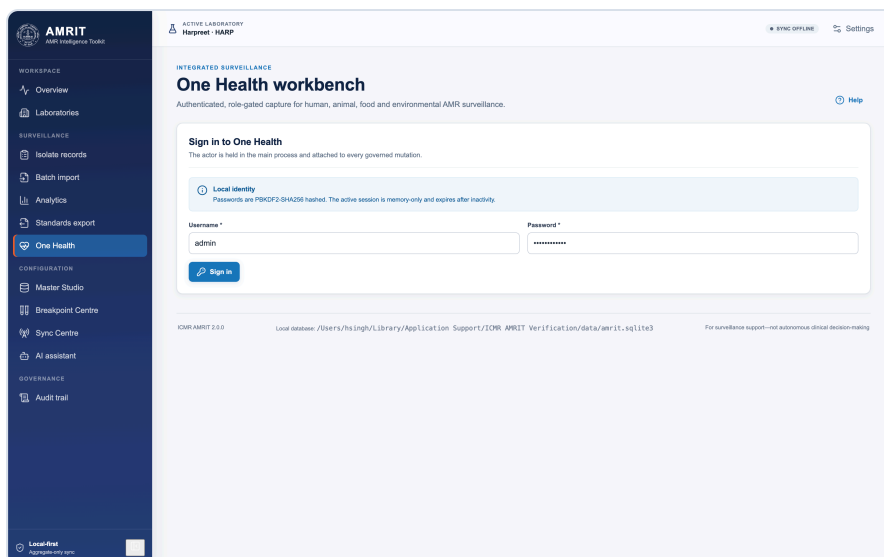
One Health ships with no identities. The first time the page is opened on a device, the sign-in panel doubles as a bootstrap form: the credentials entered create the first administrator rather than authenticate against an existing one, and the event is written to the audit trail as `one-health.user.bootstrap` with the summary “*first administrator*”. Every later identity is created from **Users** by an administrator.

Password rules are enforced in the main process

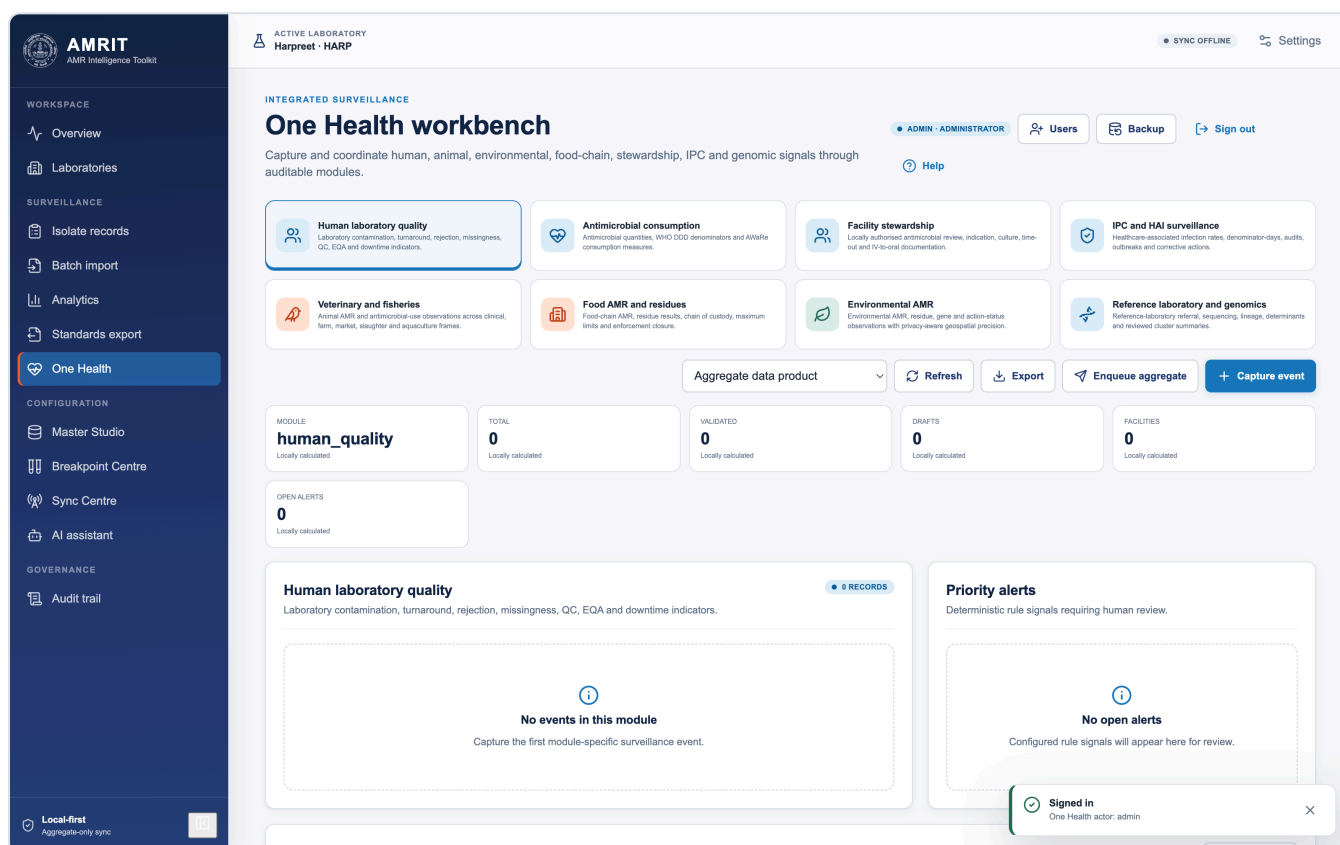
Uppercases are 3–64 characters drawn from `A–Z a–z 0–9 . _ -`. Passwords must be at least 12 characters; a shorter one is rejected before any hashing takes place, and the rejection is itself audited as `ipc.error` with the message “*Too small: expected string to have >=12 characters*”. Choose a passphrase that satisfies your institution's policy, record it in the institutional credential store, and do not reuse a clinical-system password. There is no recovery path: if the sole administrator password is lost, the One Health identities must be recreated from a backup.

12.2 Signing in

1. Open **One Health** from the Surveillance group.
2. Enter the username and password. On this reference installation the administrator account is `admin`.
3. Select **Sign in**. The signed-in actor appears beside the page title and is attached to every governed mutation. The renderer never receives a reusable credential or session token, and the session expires after 30 minutes of inactivity.



Screen 19 – Sign-in, completed. The password field is masked in the interface and is never logged. Verification is PBKDF2-HMAC-SHA256 at 310 000 iterations with a 16-byte salt, compared in constant time.



Screen 20 – One Health workbench. Eight modules, a data-product selector, module counters, the module event list, priority alerts and corrective actions. The signed-in actor badge reads **ADMIN · ADMINISTRATOR**, and **Users**, **Backup** and **Sign out** appear only for an identity that holds the matching permission.

12.3 The eight modules

MODULE	DOMAIN	EVENT TYPE	CAPTURES
Human laboratory quality	Human	human-lab-quality	Contamination, turnaround, rejection, missingness, QC, EQA and instrument downtime as numerator/denominator indicators.

MODULE	DOMAIN	EVENT TYPE	CAPTURES
Antimicrobial consumption	Human	human-amc	Antimicrobial, ATC code, AWaRe group, quantity in grams, WHO DDD, bed-days, patient-days and population.
Facility stewardship	Human	stewardship-review	Encounter reference, indication, therapy type, culture-before-antibiotic, review date, IV-to-oral eligibility, days and length of therapy, reserve authorisation.
IPC and HAI surveillance	Human	ipc-hai	CLABSI, CAUTI, VAP, SSI and other HAI types with denominator-day types, audit type and score, outbreak reference and action.
Reference laboratory and genomics	Human	reference-genomics	Referral identity, cold-chain status, sequencing, lineage, resistance determinants and reviewed cluster summaries.
Veterinary and fisheries	Animal	animal-amu-amr	Host species, production class, herd/flock/pond surrogate, sampling frame, AMR and antimicrobial-use quantities, biomass/PCU, withdrawal compliance.
Food AMR and residues	Animal	food-amr-residue	Commodity, production stage, establishment category, sampling programme, chain of custody, residue analyte, result and maximum limit, enforcement action.
Environmental AMR	Environment	environment-amr-residue	Site surrogate, matrix, sampling relation and a mandatory geospatial-precision choice from exact-restricted down to state.

EIGHT CAPTURE MODULES ACROSS THREE DOMAINS, AND THE PRODUCTS THEY FEED

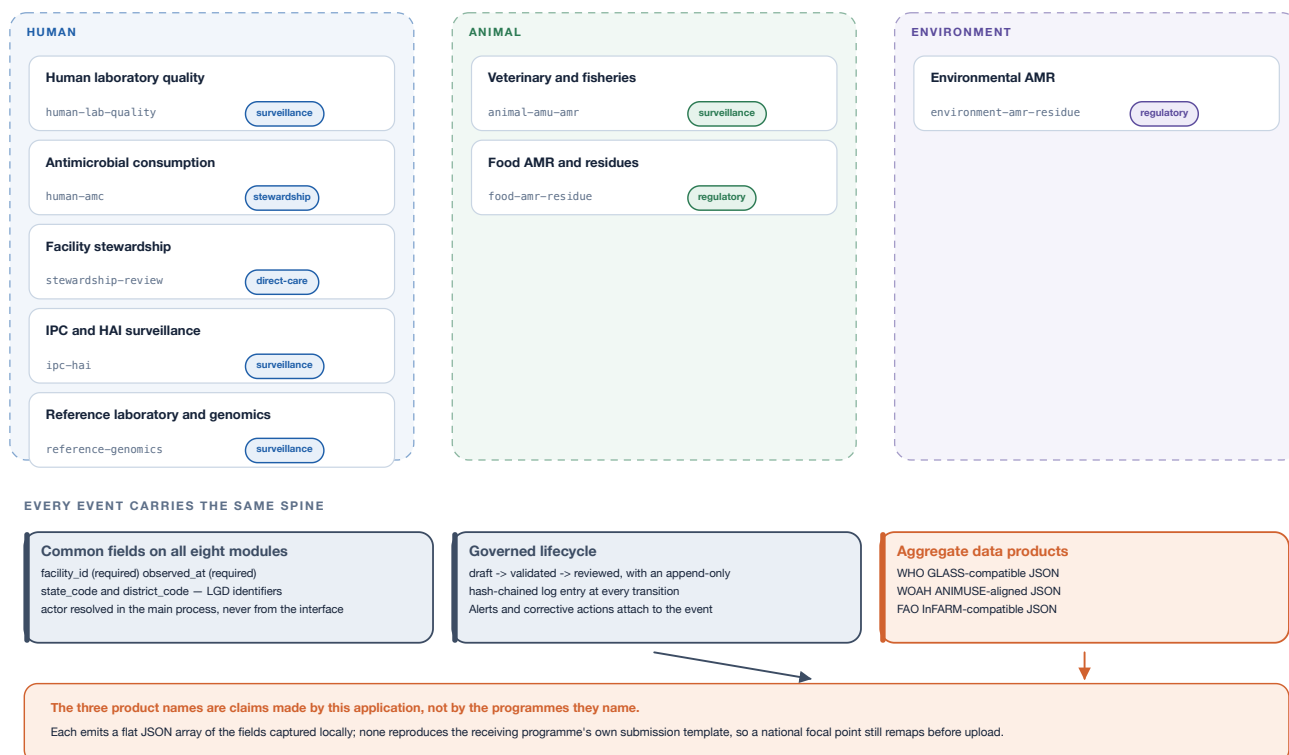
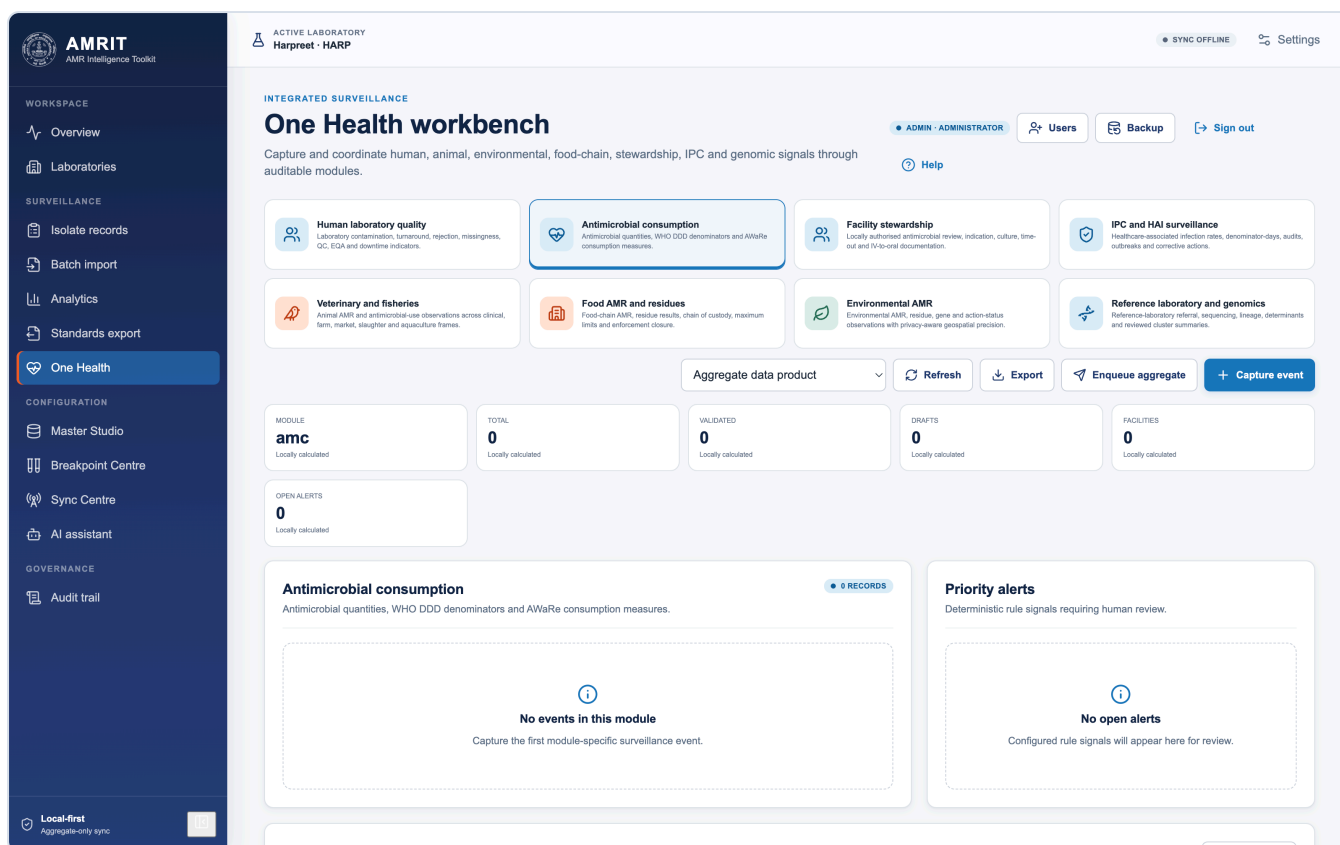


Figure 7 — One Health modules and data products. The eight capture modules across three domains, the common spine every event carries, the governed lifecycle, and the three aggregate products the workbench can emit.



Screen 21 — A module in the workbench. Selecting **Antimicrobial consumption** switches the counters, the event list and the capture form together. Every counter is labelled *locally calculated*: nothing on this page is fetched from a server.

12.4 Capturing an event

1. Select the module, then **Capture event**.
2. Complete the common spine — facility, state and district LGD codes, observation date and time — followed by the module-specific fields. Required fields are marked with an asterisk.
3. Set **Quality status** and **Sensitivity**. A new event is normally saved as **draft**; promotion to **validated** is a reviewer-authorised action.
4. Select **Validate and save**. Validation runs in the main process; a failure is reported field by field and is itself audited.

Screen 22 – Capture form, antimicrobial consumption. Facility, LGD codes and observation time are common to every module. ATC code, AWaRe group, quantity and WHO DDD are all required for this module. The dialog header restates the event type and the acting identity.

Screen 23 – The event after saving. Module counters move together — total 1, drafts 1, facilities 1 — and the event row records the acting identity and its quality status. The same write appears in the application audit trail as `one-health.capture`.

ATC is required but not validated

The ATC field is mandatory and free text. The antibiotic catalogue already carries 299 validated ATC codes — 268 human and 31 ATCvet — but the capture form does not check the entry against them, so a typographical error is accepted and flows into the exported data product. See Finding F-06.

12.5 Roles and permissions

ROLE-TO-PERMISSION MATRIX

	administrator	data-entry	reviewer	steward	auditor	sync-agent
event:create	granted	granted	—	granted	—	—
event:create:direct-care	granted	—	—	granted	—	—
event:create:regulatory	granted	—	—	granted	—	—
event:read	granted	granted	granted	granted	granted	—
event:review	granted	—	granted	granted	—	—
metrics:read	granted	—	granted	granted	granted	—
action:manage	granted	—	—	granted	—	—
audit:read	granted	—	—	—	granted	—
users:manage	granted	—	—	—	—	—
exchange:read / update	granted	—	—	granted	—	granted
health:create	granted	—	—	—	—	granted

EVENT LIFECYCLE



TAMPER-EVIDENT TRAIL AND EXCHANGE

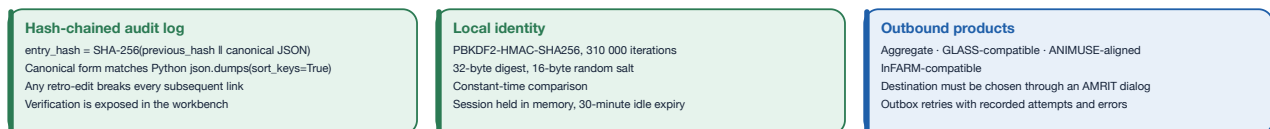
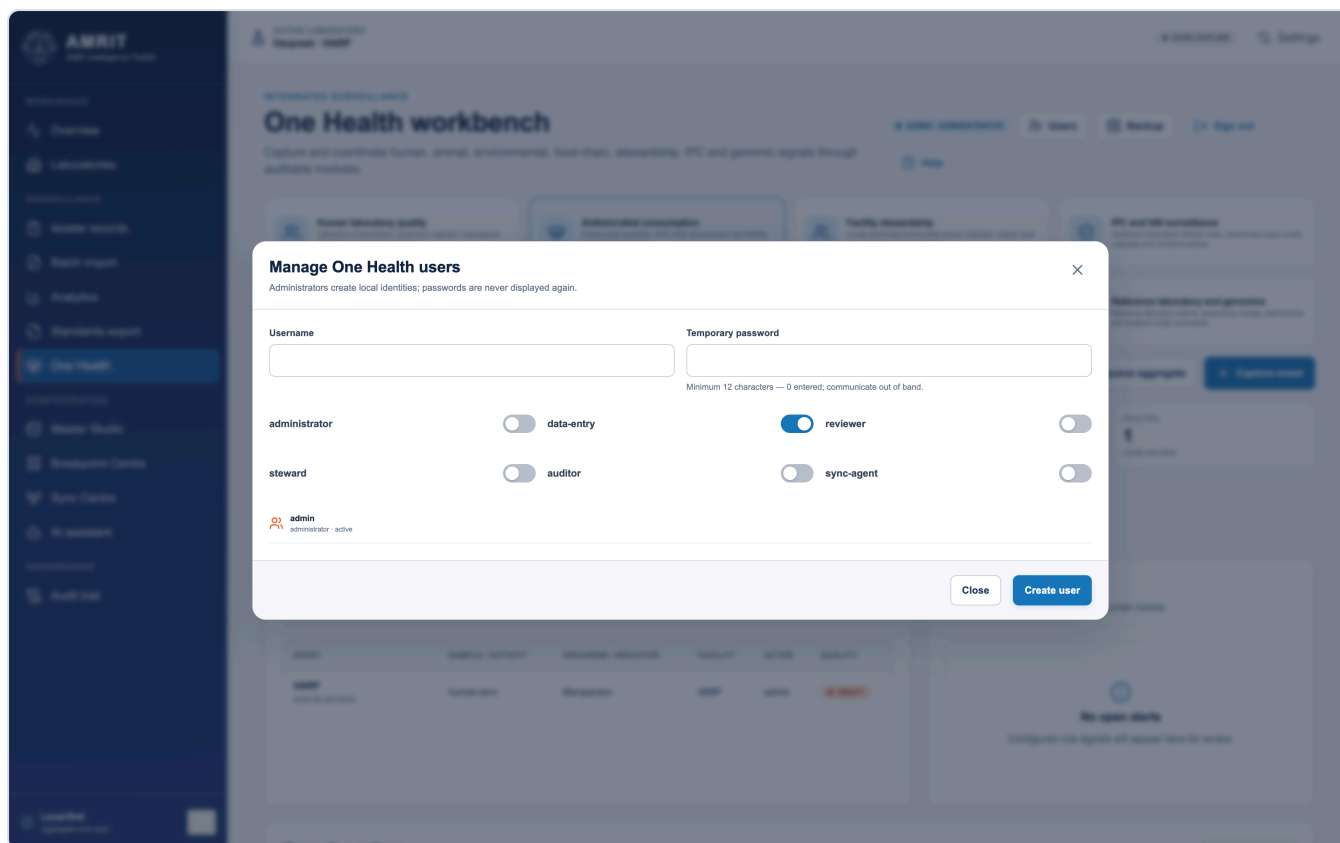


Figure 8 — One Health governance and role model. The permission matrix, the event lifecycle from capture to closure, and the assurance mechanisms behind them.

Six roles are defined — `administrator`, `data-entry`, `reviewer`, `steward`, `auditor` and `sync-agent` — and a single identity may hold several. Capture of a *direct-care* or *regulatory* event requires a second, purpose-specific permission beyond `event:create`, so a data-entry identity can record surveillance but not stewardship or enforcement.



Screen 24 — Identity administration. Reachable only with the `users:manage` permission. Roles are assigned per identity; every creation, role change and disablement is written to the hash-chained One Health log as well as the application audit trail.

12.6 Alerts, actions and closure

Deterministic rules raise alerts at medium, high or critical severity, each stamped with the ruleset version. A reviewer or steward dispositions the alert; a corrective action carries an owner, due date and priority; closure requires evidence. The result is a complete chain from observation to resolution that an auditor can follow without leaving the application.

12.7 Data products

PRODUCT	PROFILE STRING	SCOPE
Aggregate	—	Module counters and totals.
WHO GLASS-compatible	WHO-GLASS-compatible/1.0	Laboratory, date, organism, specimen, susceptibility summary, quality status.
WOAH ANIMUSE-aligned	WOAH-ANIMUSE-aligned/1.0	Species, production class, active ingredient, quantity, biomass, route, purpose.
FAO InFARM-compatible	FAO-InFARM-compatible/1.0	Animal and food sectors combined, with host or commodity, sample, organism, AST and sampling frame.

Screen 25 — Selecting a data product. The selector chooses which shape the **Export** button will produce. **Enqueue aggregate** places the same content on the national outbox for the Sync Centre instead of writing a file.

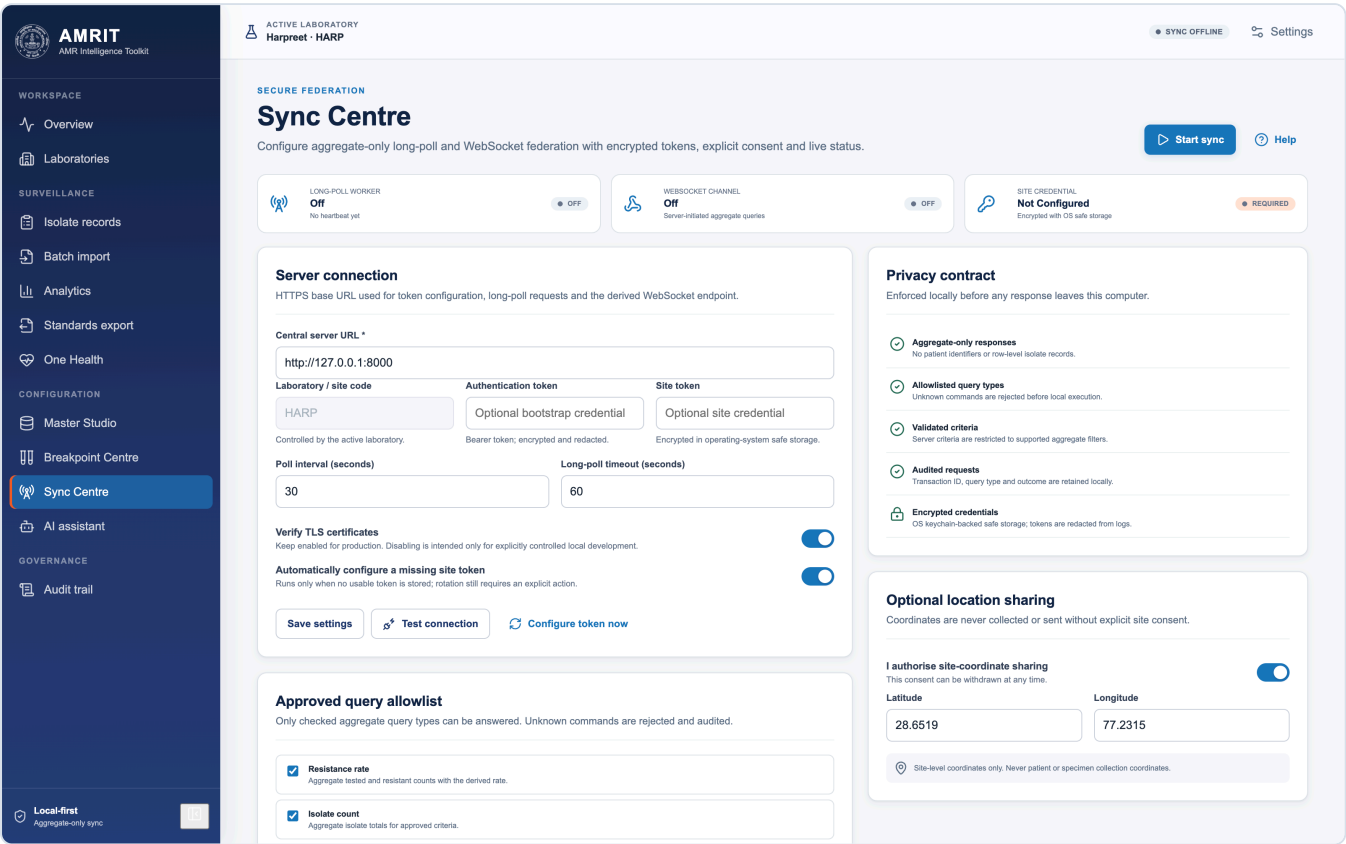
Export requires a destination chosen through an AMRIT dialog; requesting an export with an arbitrary path is refused with *“Select the destination through AMRIT first.”*

Read the profile strings as claims, not certifications

WHO-GLASS-compatible/1.0 , WOA-ANIMUSE-aligned/1.0 and FAO-InFARM-compatible/1.0 are strings this application writes about itself. None of the three reproduces the receiving programme's own submission template, and none is coded: the organism, specimen and antimicrobial travel as the free text captured locally. A national focal point must still map before upload. Finding F-10 records the position, and Chapter 23 places the work in the roadmap.

13 · Sync Centre

Federation is opt-in, aggregate-only and guarded twice. A node answers questions; it never ships records.



Screen 26 — Sync Centre. Server URL, tokens, poll interval and timeout, TLS verification, optional GPS consent and the allow-listed query types.

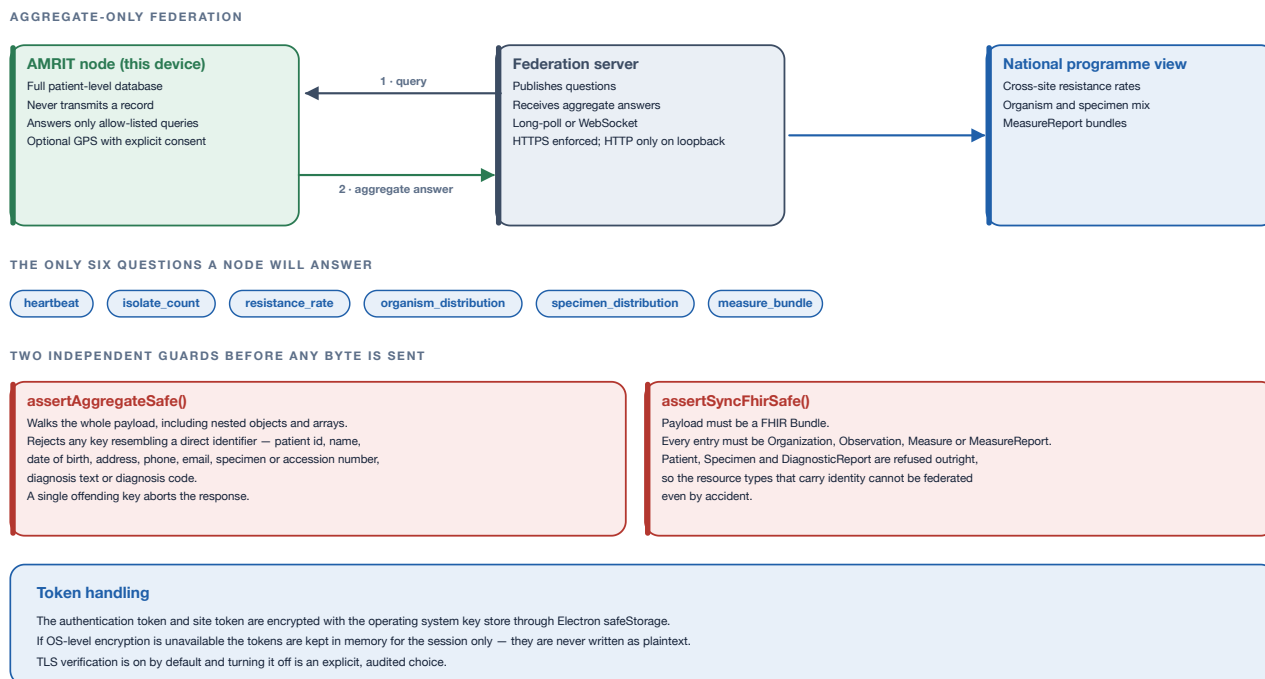


Figure 9 — Federated synchronisation. The exchange, the six answerable questions and the two guards that inspect every outbound payload.

13.1 Configuration

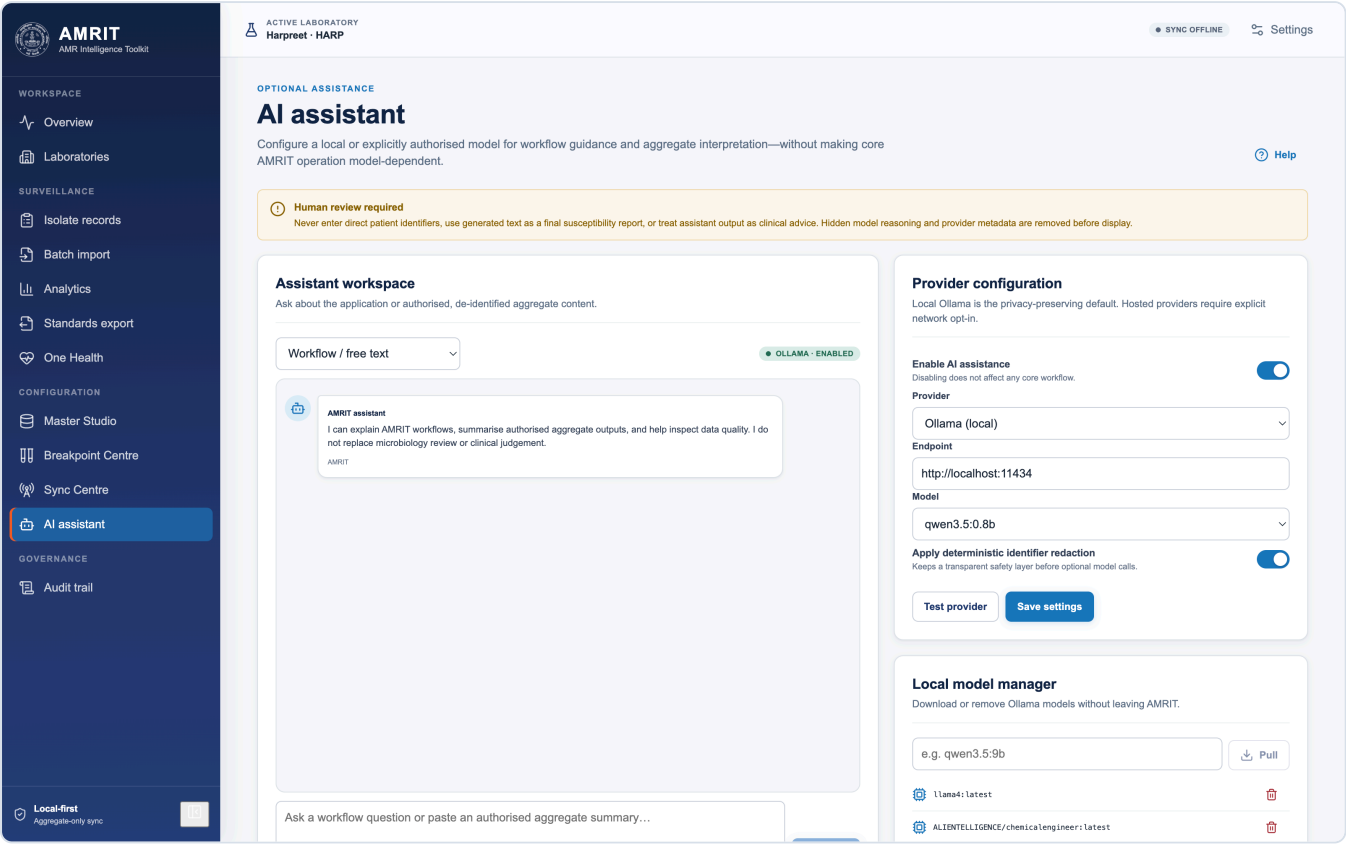
SETTING	CONSTRAINT
Server URL	Must be HTTPS. Plain HTTP is permitted only for loopback addresses.
Auth and site tokens	Encrypted with OS <code>safeStorage</code> ; memory-only if unavailable.
Poll interval / timeout	5–86 400 seconds and 5–600 seconds respectively.
Verify TLS	On by default; disabling is an explicit choice.
GPS consent	Off by default. Latitude and longitude are only transmitted when consent is on.
Allowed query types	Any subset of the six named types.

13.2 The guards

`assertAggregateSafe()` walks the entire payload including nested structures and aborts on any key resembling a direct identifier — patient identifier, name, date of birth, address, phone, email, specimen or accession number, diagnosis text or diagnosis code. `assertSyncFhirSafe()` additionally requires the payload to be a FHIR Bundle whose every entry is an Organization, Observation, Measure or MeasureReport. Patient, Specimen and DiagnosticReport are refused outright, so the resource types that carry identity cannot be federated even by mistake.

14 · AI assistant

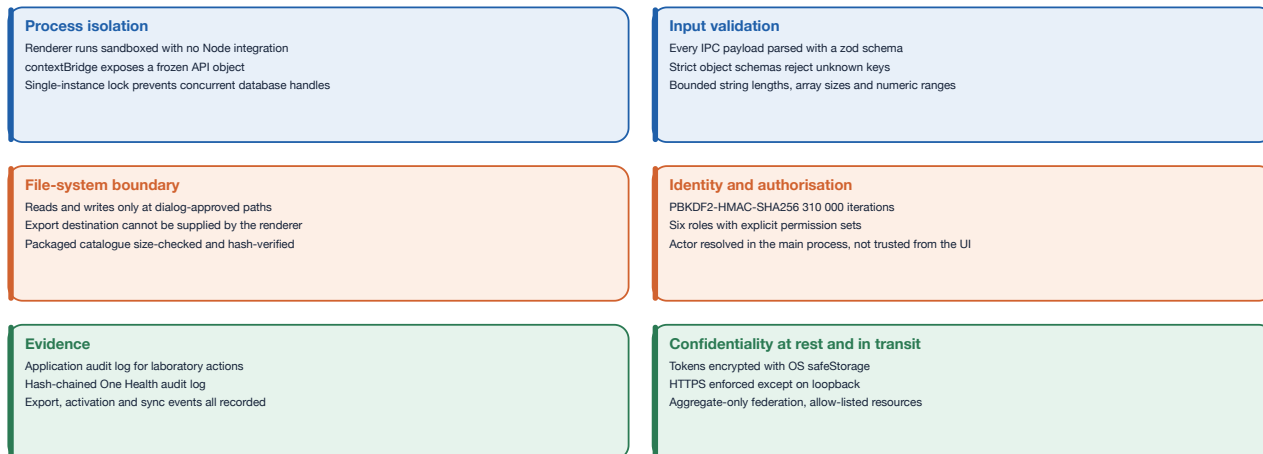
Local-first assistance for coding suggestions and narrative drafting. It is never in the decision path.



Screen 27 — AI assistant. Provider selection, model name and the prompt surface. The default provider is a local Ollama service on loopback.

14.1 Data handling

DEFENCE IN DEPTH



ASSISTANT DATA HANDLING

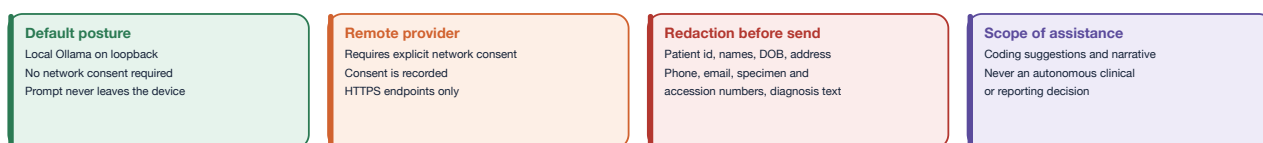


Figure 10 — Security and privacy model. The six defence layers, and the four rules that govern assistant data handling.

- The default provider is local; no network consent is required and no prompt leaves the device.
- A remote provider requires explicit network consent, which is recorded, and an HTTPS endpoint.
- Before any remote call, PHI-bearing fields are redacted: patient identifier, names, date of birth, address, phone, email, specimen and accession numbers, diagnosis and diagnosis code.

14.2 What it is used for

Built-in tasks suggest a SNOMED CT concept for a free-text organism together with the typical WHONET code, suggest a SNOMED CT specimen concept and AST panel category, and draft narrative summaries. The system prompt directs the model toward SNOMED CT, LOINC, ICD-11 and WHONET vocabulary. Output is a suggestion presented to the operator — it is never written to a record automatically.

Note for the conformance reader

The assistant's prompts are the only place in the application where ICD-11 is mentioned. There is no ICD value set, no validation and no ICD content in any export — see Finding F-08.

15 · Audit trail

Two logs, two purposes. The application audit log records laboratory actions; the One Health log is a hash chain.

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Harpreet · HARP

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Inspect traceable local operations across configuration, imports, records, exports, federation and optional AI services.

15 LOADED EVENTS

9 SUCCESSFUL

6 WARNINGS / ERRORS

Search operation, summary or details...

All outcomes

15 events shown

TIME	OPERATION	SUMMARY	OUTCOME	DETAILS
8/12/2026, 8:49:53 AM	ipc.error	one-health:capture: Indicator is required; Numerator is required; Denominator is required	ERROR	Inspect
8/12/2026, 8:48:54 AM	one-health.user.bootstrap	admin: first administrator	OK	Inspect
8/12/2026, 8:40:56 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:48 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:29 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:23 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 7:03:58 AM	catalogue.genomic-markers	52 markers (v1-2026-08)	OK	Inspect
8/12/2026, 4:20:06 AM	sync.settings	Sync settings saved without plaintext credentials.	OK	Inspect
8/12/2026, 4:18:08 AM	master.domains.activation	veterinary deactivated	OK	Inspect
8/12/2026, 4:18:00 AM	ipc.error	masters:delete: One Health domains record veterinary is used by 1 record(s); deactivate it instead.	ERROR	Inspect
8/12/2026, 4:17:52 AM	master.domains.activation	animal deactivated	OK	Inspect

Screen 28 — Audit trail. Searchable action history with actor, outcome, detail and timestamp. Failures are recorded as prominently as successes.

15.1 What is recorded

EVENT CLASS	EXAMPLES
Record lifecycle	Isolate saved, duplicated, deleted; import run committed with row counts.
Configuration	Master entry created, edited, disabled; laboratory cloned; catalogue seeded with row counts and version.
Breakpoints	Workbook staged with its hash; set activated with the actor.
Release	Export written with format, record count, laboratory and file name; sync exchange outcome.
One Health	User bootstrap, sign-in, event capture, alert review, action closure — each additionally hash-chained.
Failures	Rejected IPC input, validation errors and refused operations, with the reason.

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Inspect traceable local operations across configuration, imports, records, exports, federation and optional AI services.

16 LOADED EVENTS

10 SUCCESSFUL

6 WARNINGS / ERRORS

Search operation, summary or details...

All outcomes

16 events shown

TIME	OPERATION	SUMMARY	OUTCOME	DETAILS
8/12/2026, 9:43:16 AM	one-health.capture	amc: b5570d95-ebb4-433c-ae4a-0ec97b55a1de	OK	Inspect
8/12/2026, 8:49:53 AM	ipc.error	one-health.capture: Indicator is required; Numerator is required; Denominator is required	ERROR	Inspect
8/12/2026, 8:48:54 AM	one-health.user.bootstrap	admin: first administrator	OK	Inspect
8/12/2026, 8:40:56 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:48 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:29 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 8:40:23 AM	ipc.error	one-health.bootstrap-admin: input: Too small: expected string to have >=12 characters	ERROR	Inspect
8/12/2026, 7:03:58 AM	catalogue.genomic-markers	52 markers (v1-2026-08)	OK	Inspect
8/12/2026, 4:20:06 AM	sync.settings	Sync settings saved without plaintext credentials.	OK	Inspect
8/12/2026, 4:18:08 AM	master.domains.activation	veterinary deactivated	OK	Inspect
8/12/2026, 4:18:00 AM	ipc.error	masters.delete: One Health domains record veterinary is used by 1 record(s); deactivate it instead.	ERROR	Inspect

Screen 29 — The audit trail immediately after a governed write. The One Health capture from Screen 17 appears as `one-health.capture` with the module key and the event identifier, above the earlier `one-health.user.bootstrap` entry. The four `ipc.error` rows are rejected inputs — a password below the 12-character minimum and an incomplete capture — and they are retained deliberately: a log that records only successes cannot be used to investigate anything.

15.2 The hash chain

Each One Health audit entry stores `entry_hash = SHA-256(previous_hash || canonical_json(entry))`. The canonical form matches Python's `json.dumps(sort_keys=True, separators=(",", ":"), ensure_ascii=True)`, so an independent verifier written in another language reproduces the same digest. Altering or removing an entry breaks every subsequent link, and verification is exposed in the workbench.

16 · Process architecture

AMRIT is an Electron application with a hard boundary between the sandboxed interface and the privileged main process. Everything that touches the database, the file system or the network runs on the privileged side and is validated at the boundary.

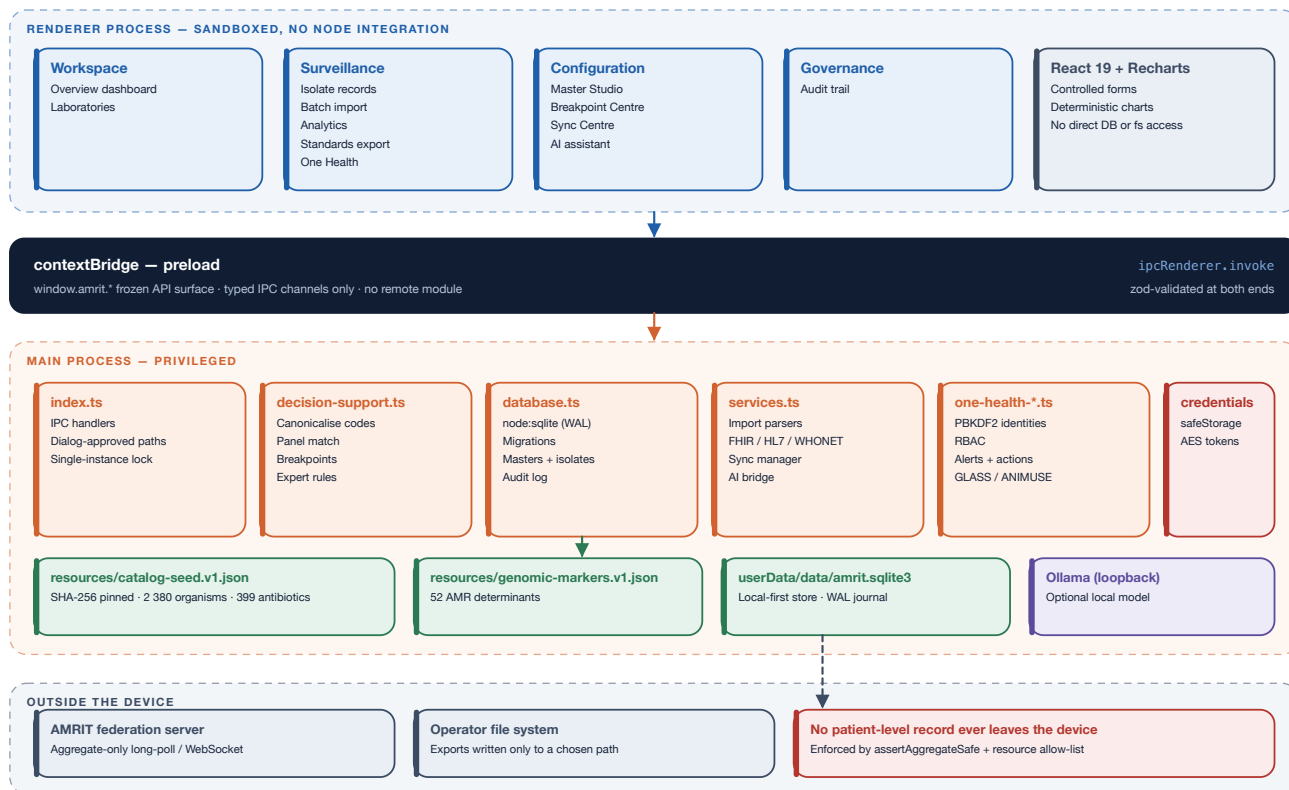


Figure 11 — AMRIT process architecture. Renderer, preload bridge, main-process modules, packaged resources and the three things that exist outside the device.

16.1 The boundary

- The renderer runs sandboxed with no Node integration and no remote module.
- `contextBridge.exposeInMainWorld` publishes a **frozen** API object. The renderer cannot monkey-patch it, and it exposes typed channels rather than arbitrary capability.
- Every IPC payload is parsed with a zod schema on arrival. Object schemas are strict, so an unexpected key is an error rather than an ignored field.
- A single-instance lock prevents two application instances from opening the same database file.

16.2 Main-process modules

MODULE	LINES	RESPONSIBILITY
database.ts	3 382	Schema, migrations, master definitions, records, audit, aggregate queries.
services.ts	2 107	Import parsing, all standards exports, sync manager, assistant bridge.
decision-support.ts	827	Canonicalisation, panel matching, breakpoint interpretation, expert rules.
index.ts	704	IPC handlers, dialog-approved paths, window lifecycle.

MODULE	LINES	RESPONSIBILITY
analysis-engine.ts	539	The ten analysis modes.
whonet-legacy.ts	476	WHONET configuration and library ingestion, WHO AWaRe groups.
one-health-*.ts	537	Module definitions, governance, RBAC, data products.

16.3 Quality gates

`pnpm run check` runs the TypeScript compiler and ESLint with `--max-warnings=0`. `pnpm test` runs the Vitest suite; at the time of this verification it reported **100 passed and 2 skipped across 18 files**, with one file skipped in full. `pnpm run dist` runs both before it will produce an installer.

17 · End-to-end data flow

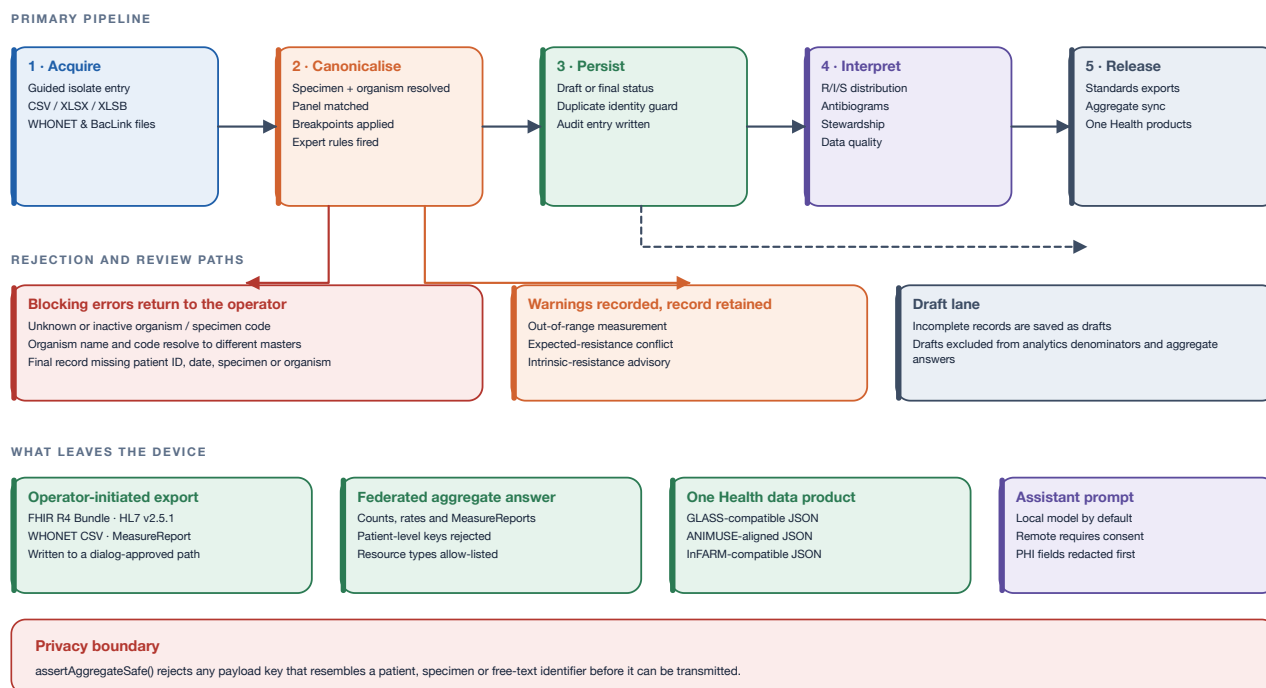


Figure 12 — End-to-end data flow. The five-stage pipeline, the rejection and review paths that branch from it, and the four channels through which anything leaves the device.

17.1 Reading the diagram

The top row is the happy path: acquire, canonicalise, persist, interpret, release. The middle band shows where records leave that path — blocking errors return to the operator, warnings are recorded without discarding the record, and incomplete records fall into the draft lane where they are visible but excluded from denominators.

The bottom band is the only set of exits from the device, and each is operator-initiated: a standards export to a chosen file, an aggregate answer to a configured federation server, a One Health data product to a chosen file, and — if a remote provider has been consented to — a redacted assistant prompt.

18 · Security and privacy model

Six layers, each of which assumes the layer above it may fail.

LAYER	CONTROL
Process isolation	Sandboxed renderer, no Node integration, frozen contextBridge surface, single-instance lock.
Input validation	zod schemas on every channel; strict objects; bounded strings, arrays and numbers; URL scheme enforcement.
File-system boundary	Reads and writes only at dialog-approved paths; packaged resources size-checked and hash-verified before parsing.
Identity and authorisation	PBKDF2-HMAC-SHA256 at 310 000 iterations with a 16-byte salt and 32-byte digest, compared in constant time; six roles with explicit permission sets; the actor is resolved in the main process and never trusted from the interface.
Evidence	Application audit log plus a hash-chained One Health log.
Confidentiality	Tokens encrypted with OS <code>safeStorage</code> ; HTTPS enforced off-loopback; aggregate-only federation with a resource allow-list; PHI redaction before any remote assistant call.

Hardening observation

310 000 PBKDF2-SHA256 iterations meets the older OWASP guidance but sits below the current recommendation of 600 000 for PBKDF2-HMAC-SHA256. The stored hash format already encodes its iteration count and the verifier accepts 100 000–2 000 000, so the parameter can be raised for new and re-set passwords without a migration. This is recorded as Finding F-12.

19 · Verification method and scope

This chapter states exactly what was tested, how, and against what. Every finding in Chapter 22 is reproducible from the steps in Appendix B.

19.1 What was verified

Application	ICMR AMRIT 2.0.0, compiled with <code>electron-vite</code> build from the working tree
Catalogue	<code>icmr-amrit-packaged-catalogue</code> version 2026.1, SHA-256 <code>636aa336...7730f10</code>
Live data	A copy of the operator database: 11 100 isolates across 5 laboratories
Isolation	Run against a sandboxed database copy with a private user-data directory, so no production data was touched
Regression suite	<code>pnpm run check clean; pnpm test</code> — 100 passed, 2 skipped across 18 files
Verification date	12 August 2026, against a build compiled the same day

This chapter records a second, independent pass

The conformance position in Chapters 20 to 24 was re-derived from the compiled build on 12 August 2026 rather than carried forward. The catalogue was re-counted, the export builders were re-read, fresh FHIR, HL7, WHONET and MeasureReport artefacts were produced from the live snapshot, and the interface was re-driven end to end including a One Health sign-in and capture. Every coverage figure and every quoted artefact in this part comes from that run. Two numbers reported in the earlier pass were corrected in the process, and one rating was lowered; both changes are marked where they occur.

19.2 Method

- Catalogue measurement.** Every terminology-bearing column in the packaged catalogue was counted for non-empty values, giving the coverage percentages in Chapter 20.
- Source inspection.** The export builders, the decision-support canonicalisation, the aggregate-query builder and the One Health data products were read in full and their terminology behaviour traced to source lines.
- Artefact production — synthetic.** A fresh database was seeded from the packaged catalogue, a single isolate was saved through the ordinary save path, and FHIR, HL7, WHONET CSV, MeasureReport and aggregate artefacts were produced from it. This isolates what the *application* does, free of any historical data quirk.
- Artefact production — live.** The same exports were produced from a snapshot of the operator database, to confirm the behaviour under real, historically accumulated data.
- Interactive verification.** The compiled build was driven through the interface — including a One Health sign-in and a real antimicrobial-consumption capture — and the resulting database rows were read back directly.

19.3 Standards assessed

FAMILY	STANDARDS
Exchange formats	HL7 FHIR R4; HL7 v2.5.1 (ORU^R01, batch protocol); WHONET/BacLink
Clinical terminology	SNOMED CT; LOINC; ICD-10 and ICD-11

FAMILY	STANDARDS
Medicinal product and use	WHO ATC; WHO DDD; WHO AWaRe (2025 edition)
Units and quantities	UCUM
Taxonomy	GBIF / NCBI taxonomy and Linnaean rank chain
Geography and time	ISO 3166-1; ISO 8601; India LGD codes (Ministry of Panchayati Raj)
Surveillance programmes	WHO GLASS; WOAAH ANIMUSE; FAO InFARM; ICMR AMRSN alignment
National digital health	ABDM / NDHM identifier and document conventions

19.4 Scoring vocabulary

RATING	MEANING
Conformant	Implemented as the standard requires, and demonstrated in a produced artefact.
Partial	Implemented in part; a receiver gets usable but incomplete or non-validating content.
Held, not emitted	The standard code is present in the application's own data but never appears in any output.
Structural only	The output has the shape the programme expects, but its content is uncoded free text.
Not implemented	No binding, no value set and no output.

20 • Terminology inventory and coverage

AMRIT's terminology assets are stronger than its exports suggest. This is the measured inventory of what the application actually holds.

20.1 Coverage of the packaged catalogue

CATALOGUE COLUMN	STANDARD	POPULATED	COVERAGE	EXAMPLE
organisms.snomed_code	SNOMED CT	2 099 / 2 380	88.2 %	112283007 – Escherichia coli
organisms.concept_code	SNOMED CT	2 099 / 2 380	88.2 %	equal to snomed_code except 2 rows
organisms.system	URI	2 099 / 2 380	88.2 %	http://snomed.info/sct
organisms.gbif_taxon_id	GBIF	2 068 / 2 380	86.9 %	6110934
organisms.genus_name	Taxonomy	2 087 / 2 380	87.7 %	Escherichia
organisms.taxonomic_status	Taxonomy	2 380 / 2 380	100 %	accepted / synonym
antibiotics.atc_code	WHO ATC / ATCvet	299 / 399	74.9 %	268 human J01MA02 • 31 vet QJ01MA92
antibiotics.who_aware	WHO AWaRe	191 / 399	47.9 %	Access 49 • Watch 120 • Reserve 22
antibiotics.loinc_mlc	LOINC (MIC)	121 / 399	30.3 %	184–2 – Ciprofloxacin MIC
antibiotics.loinc_sbt	LOINC (disk)	117 / 399	29.3 %	187–5 – Ciprofloxacin disk
antibiotics.din_code	DIN	214 / 399	53.6 %	AMC
antibiotics.jac_code	JAC	169 / 399	42.4 %	AMPC–CVA
antibiotics.eucast_code	EUCAST	0 / 399	0 %	column present, unpopulated
samples.concept_code	SNOMED CT	3 / 8	37.5 %	122575003 – Urine
states.lgd_code	India LGD	36 / 36	100 %	32 – Kerala
districts.lgd_code	India LGD	785 / 785	100 %	554 – Alappuzha (state 32)
codeValues country	ISO 3166-1	252 / 252	100 %	AFG + iso2 AF + WHO region EMRO
genomic markers	Gene symbol only	52 / 52	100 %	blaNDM – no NCBI RefGene or CARD ARO id

Correction to the earlier pass: the LGD row previously read “36 / 785”, conflating two separate catalogues. They are counted separately above. The `concept_code` row previously read “identical to `snomed_code`”; re-measurement found two organisms — *Klebsiella pneumoniae* and *Staphylococcus aureus* — where the two columns hold different SNOMED concepts, so a consumer must be told which column is authoritative.

This is a genuine strength

A packaged catalogue of 2 380 organisms with 88 % SNOMED CT coverage, full taxonomic rank chains, WHO ATC on three quarters of the antibiotic list, LOINC susceptibility codes, AWARe classification, 785 LGD district codes and ISO 3166 country codes with WHO regions is a substantially better terminology foundation than most laboratory surveillance tools ship with. The content-integrity controls around it — pinned SHA-256, per-collection minimums, referential integrity checks and a PII scan — are stronger still.

20.2 Where the terminology travels

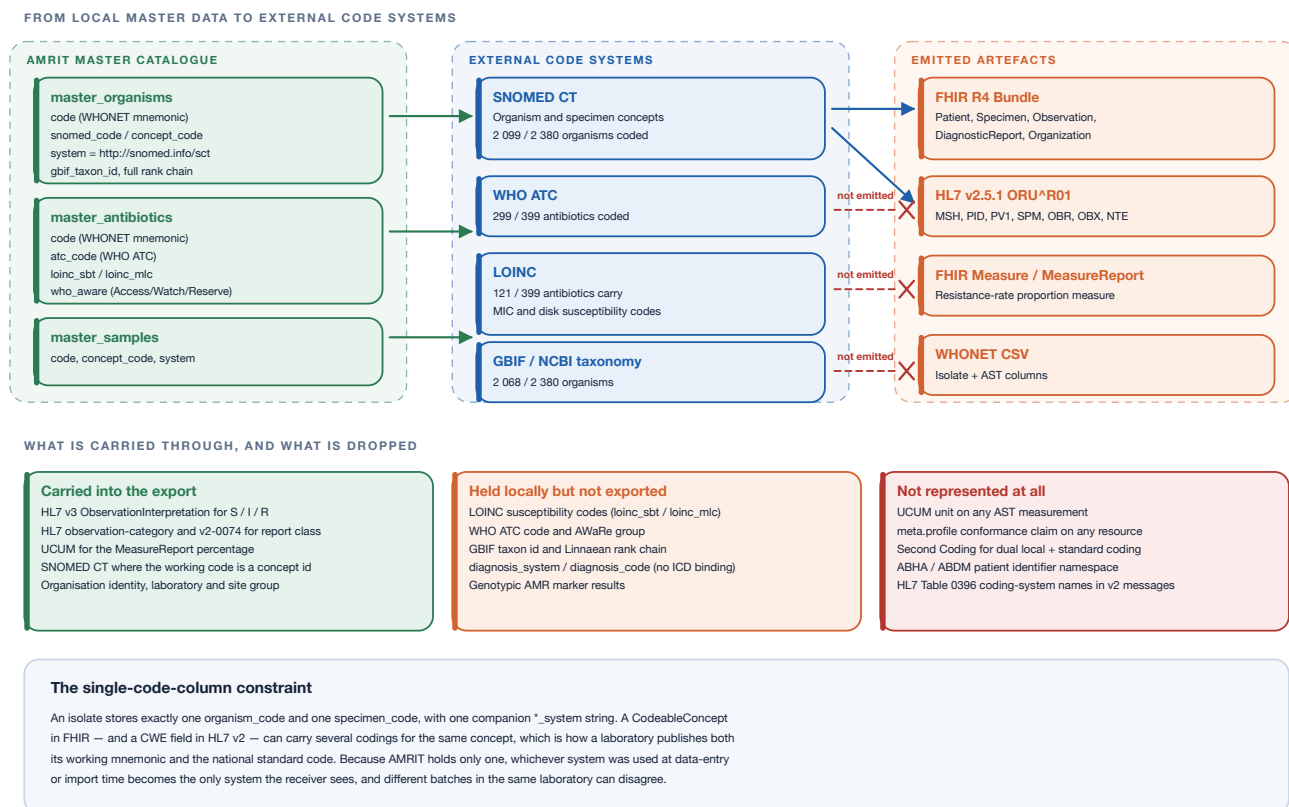


Figure 13 — Terminology and interoperability map. The crossed connectors are the finding: WHO ATC, LOINC and taxonomy identifiers are held but never reach an emitted artefact.

20.3 The structural constraint behind most findings

An isolate record stores exactly one organism code and one specimen code, each with a single companion *_system string:

```
isolates(... specimen_system TEXT, specimen_code TEXT,
           organism TEXT, organism_system TEXT, organism_code TEXT,
           diagnosis_system TEXT, diagnosis_code TEXT, ...)
```

A FHIR `CodeableConcept` — and a CWE field in HL7 v2 — is explicitly designed to carry *several* codings for one concept. That is how a laboratory publishes its working mnemonic *and* the national standard code in the same message, letting each receiver pick the system it understands. Because AMRIT stores only one code per concept, whichever system happened to be used at entry or import becomes the only system a receiver ever sees.

21 · Conformance matrix

CONFORMANCE STATUS BY STANDARD

STANDARD	STATUS	EVIDENCE
HL7 FHIR R4 (structure)	Conformant	Bundle, entry.fullUri and RFC 4122 v5 resource ids; references resolve within the bundle.
HL7 FHIR R4 (terminology binding)	Partial	HL7 code systems used for interpretation, category and measure scoring; no LOINC on any observation.
HL7 v2.5.1 ORU^R01	Partial	Correct segment order and escaping; MSH-9 lacks the message-structure component and coding systems are URIs.
SNOMED CT	Partial	2 099 of 2 380 organisms and 3 of 8 specimens carry concept ids; emission depends on the working code.
LOINC	Held, not emitted	121 antibiotics carry MIC and disk susceptibility codes in the catalogue; no export references them.
UCUM	Partial	Used for the MeasureReport percentage; absent from every MIC and zone-diameter value.
WHO ATC	Held, not emitted	299 of 399 antibiotics coded; required free-text field in One Health AMC, not validated against the catalogue.
WHO AWaRe (2025)	Conformant	191 antibiotics classified; Access / Watch / Reserve enforced as a choice list in stewardship capture.
ICD-10 / ICD-11	Not implemented	diagnosis_code and diagnosis_system columns exist and accept free text; no value set and no export.
ISO 3166-1	Conformant	Alpha-3 country codes with alpha-2 and WHO region held as metadata.
ISO 8601	Partial	Dates normalised on isolate records; One Health observed_at stores a local datetime with no seconds or offset.
India LGD codes (MoPR)	Conformant	36 states and union territories and 785 districts; bound in laboratory setup and One Health capture.
WHONET / BacLink	Partial	Native code space and complete CSV content; header row uses prose labels, not the 425 WHONET field names shipped.
WHO GLASS	Structural only	Self-declared compatible JSON; organism, specimen and susceptibility pass through as free text.
WOAH ANIMUSE	Structural only	Self-declared aligned JSON; country fixed to IND, species and ingredient uncoded.
FAO InFARM	Structural only	Self-declared compatible JSON; commodity and sampling frame uncoded.
ABDM / NDHM (India)	Not implemented	No ABHA identifier namespace, no document Bundle with a Composition, no meta.profile claim.

Figure 14 — Standards conformance summary. Seventeen standards assessed against the compiled 2.0.0 build.

WHAT AMRIT IS MEASURED AGAINST, AND WHERE IT SITS TODAY



A green dot means the standard is met for the data the application holds. An amber dot means it is met for part of that data. A red dot means a receiving system cannot rely on it.

Figure 15 — National and international standards landscape. The same assessment grouped by the authority that publishes each standard, with the Indian national row separated out. A green dot means the standard is met for the data the application holds; amber, for part of it; red, that a receiving system cannot rely on it.

21.1 Summary

RATING	COUNT	STANDARDS
Conformant	4	FHIR R4 structure · WHO AWaRe · ISO 3166-1 · India LGD / MDDS geography
Partial	6	FHIR R4 terminology binding · HL7 v2.5.1 · SNOMED CT · UCUM · ISO 8601 · WHONET/BacLink
Held, not emitted	2	LOINC · WHO ATC / ATCvet
Structural only	3	WHO GLASS · WOAH ANIMUSE · FAO InFARM
Not implemented	2	ICD-10 / ICD-11 · ABDM / NDHM

Change from the earlier pass: WHONET/BacLink is lowered from **Conformant** to **Partial**. The exported CSV carries the right content, but its header row uses prose labels — `Patient ID`, `Specimen number`, `Organism code` — where WHONET's own field dictionary uses `PATIENT_ID`, `SPEC_NUM` and `ORGANISM`. The application already ships that dictionary: 425 distinct WHONET field names across 898 field definitions, including all 14 of the core identifiers. A file whose headers do not match cannot be loaded by the receiving software without a mapping step, which is the definition of partial rather than full conformance.

21.2 Interoperability by receiving system

The practical question is not “is it standards-based” but “what can a given receiver actually do with the output”. Assessed that way:

RECEIVER	USABLE TODAY	QUALIFICATION
A WHONET-based national node	With work	The CSV carries result, measurement, method, guideline and potency per antibiotic — the content is complete. Column headers are human-readable labels rather than WHONET field names, so a column mapping must be defined once per exporting site before the file will load (F-13).
A FHIR server ingesting lab results	With work	The bundle loads and its references resolve. Observations cannot be found by LOINC code, and quantitative values have no unit, so automated interpretation needs a local adapter.
An HL7 v2 laboratory interface	With work	Segments and escaping are correct. The interface engine must tolerate URI values in the coding-system component and a two-component MSH-9, and must know that OBX-6 will be empty.
A programme aggregating resistance rates	Yes	The MeasureReport is the strongest artefact: a proportion measure with populations, a UCUM percentage and an organism stratifier.
WHO GLASS-AMR submission	Not directly	The JSON is a locally defined shape with uncoded content; it is a staging product for a submission tool, not a submission.
An ABDM-connected health record	No	No ABHA identifier namespace, no document Bundle with a Composition, no profile claim.

22 · Findings

Fourteen findings, each with the source location that causes it and the observed output that demonstrates it. Severity reflects the consequence for a receiving system, not the effort to fix.

F-01 · A code system is asserted that does not match the code

High

WHERE `decision-support.ts:361-362` and `:392-393`; consumed at `services.ts:846`, `:864`, `database.ts:2524`

CAUSE Canonicalisation writes the master's **working code** into `*_code` and the master's **code system URI** into `*_system`. For a master row whose `system` is `http://snomed.info/sct` but whose `code` is a WHONET mnemonic, the pair asserts a SNOMED CT code that does not exist.

OBSERVED Freshly seeded catalogue, one isolate saved through the normal path:

```
"type": { "coding": [ { "system": "http://snomed.info/sct",
                        "code": "URINE",           <-- not a SNOMED CT concept id
                        "display": "Urine" } ] }

"valueCodeableConcept": { "coding": [ { "system": "http://snomed.info/sct",
                                         "code": "ECO",      <-- WHONET mnemonic
                                         "display": "Escherichia coli" } ] }
```

The catalogue holds the correct concepts — `122575003` for urine and `112283007` for *E. coli* — in `concept_code`, which the export never reads. The same mismatch reaches HL7 (SPM-4, OBX-5) and the federated aggregate codings.

SCALE This is not an edge case. Counted directly against the operator database on 12 August 2026:

```
master_organisms  system = SNOMED CT and code NOT numeric -> 2 094 of 2 099
master_samples    system = SNOMED CT and code NOT numeric ->    3 of 3
```

Every one of those 2 097 master rows produces a false SNOMED CT assertion the moment a record canonicalised against it is exported. The five organism rows that would be emitted correctly are those whose working code happens to be the concept id itself.

IMPACT A receiver performing a SNOMED CT lookup fails outright, or silently records an unresolvable code. FHIR terminology validation rejects the resource.

COMPOUNDING In the live database the same column holds three different systems across rows of one laboratory:

```
organism_code  numeric + system SNOMED CT           10 000 rows
organism_code  alphanumeric + system empty           1 100 rows
specimen_code  numeric + system SNOMED CT           10 089 rows
specimen_code  alphanumeric + system https://amrit.local/specimen 1 000 rows
specimen_code  alphanumeric + system empty           11 rows
```

The 10 000 correctly coded rows were loaded directly, not saved through the interface. Any record that is saved through the interface acquires the mismatched pair described above, so the proportion of affected rows grows with ordinary use. `https://amrit.local/specimen` is also a non-resolvable namespace that no external receiver can dereference.

REMEDY Emit `concept_code` with the standard system and the working code as a second `Coding` under an ICMR-owned CodeSystem URI. Never pair a standard system URI with a non-standard code.

F-02 · No LOINC code on any observation

High

WHERE `services.ts:858` (organism identification), `:879` (susceptibility), `:895` (report), `:1018` and `:1027` (HL7 OBX-3)

CAUSE Observation codes are either free text or a locally invented system.

OBSERVED

```
"code": { "text": "Organism identified" }           <-- no coding at all
"code": { "coding": [ { "system": "urn:whonet:antibiotic-code",
                        "code": "CIP", "display": "Ciprofloxacin" } ],
  "text": "CIP susceptibility" }
"code": { "text": "Microbiology culture and antimicrobial susceptibility" }
```

The antibiotic catalogue already holds the LOINC codes for this test: `CIP` → `loinc_mlc` 184–2 (MIC) and `loinc_sbt` 187–5 (disk diffusion), for 121 and 117 agents respectively. Neither is referenced by any export.

IMPACT LOINC is how a laboratory result is identified in FHIR and HL7 v2. Without it a receiver cannot query, route or trend the observation without bespoke mapping of the `urn:whonet:` namespace, which is not a registered code system.

REMEDY Set `Observation.code` to the LOINC code selected by test method — `loinc_mlc` for MIC, `loinc_sbt` for disk diffusion — with the WHONET code retained as a second Coding. Use LOINC 634–6 for the organism-identification observation and a LOINC microbiology-study code on the DiagnosticReport.

F-03 · Quantitative susceptibility results are exported without units

High

WHERE `services.ts:885` (FHIR `valueQuantity`); `services.ts:1027` (HL7 OBX-6 left empty)

OBSERVED From the live database — a disk zone and an MIC are indistinguishable:

```
"code": {"coding": [{"system": "urn:whonet:antibiotic-code", "code": "CIP"}]},
"interpretation": [ ... "code": "R" ... ],
"valueQuantity": { "value": 10 }           <-- 10 mm zone? 10 mg/L MIC? unknowable
```

```
OBX|5|NM|CIP^Ciprofloxacin susceptibility^WHONET||10||R||F||20260710
      ^^^ OBX-6 Units is empty
```

IMPACT The highest-consequence finding in this report. A 10 mm zone means resistant; a 10 mg/L MIC means something entirely different for a different agent. A receiver that trends these values without knowing the method will produce wrong numbers, and the method is not carried in the FHIR observation at all.

NOTE The information exists: the AST result stores `method` (“MIC”, “Disk diffusion”) alongside the measurement, and the isolate editor displays it. It is dropped at the export boundary.

REMEDY Populate `valueQuantity.unit / system / code` with UCUM `mg/L` for MIC and `mm` for disk diffusion, add the method as `Observation.method`, and set HL7 OBX-6 to the same UCUM unit. Where an MIC is censored, emit `valueQuantity.comparator` rather than a bare number.

F-04 · HL7 v2 coding-system components carry URIs, not table 0396 names

Medium

WHERE `services.ts:1016` (SPM-4), `:1018` (OBX-5), `:1027` (OBX-3)

OBSERVED

```
SPM|1|SYN-HARP-009992|SYN-HARP-009992|119334006^Sputum^http://snomed.info/sct|||20260710
OBX|1|CWE|ORG^Organism identified^L||56415008^Escherichia coli^http://snomed.info/sct|||||F
OBX|2|NM|AMC^Amoxicillin/Clavulanic acid susceptibility^WHONET||27|||S|||F|||20260710
```

The third component of a CWE is the *name of coding system*, drawn from HL7 Table 0396 — `SCT` for SNOMED CT, `LN` for LOINC, and a `99`-prefixed local namespace for anything site-defined. A URI is not a valid value, and neither `L` nor `WHONET` is a registered name.

IMPACT Strict interface engines reject the segment; permissive ones treat the coding system as unknown and fall back to the display text.

REMEDY Map system URIs to Table 0396 names, and register local namespaces as `99AMRIT` / `99WHON`.

F-05 · HL7 v2.5.1 header and batch envelope are incomplete

Medium

WHERE `services.ts:1006–1037`

OBSERVED

```
MSH|^~\&|AMRIT|HARP|INTEROP|FHIR|20260812033646||ORU^R01|HARP-11593-...|P|2.5.1
      ^^^^^^^^^^          ^^^^^^
      placeholder routing  MSH-9 has no third component
```

- **MSH-9** should be `ORU^R01^ORU_R01` in v2.5.1; the message-structure component is absent.
- **MSH-3/4/5/6** — receiving application and facility are hardcoded to `INTEROP` and `FHIR`, which carry no routing meaning, and no namespace or universal ID is supplied.
- **MSH-18** (character set) is absent; UTF-8 content should declare `UNICODE UTF-8`.
- **MSH-21** (message profile identifier) is absent, so the message asserts no conformance profile.
- **Batch protocol** — messages are concatenated with a newline. A true HL7 batch requires `FHS` / `BHS` ... `BTS` / `FTS` framing.

POSITIVE Segment terminators are correctly `\r`, escaping of `\ | ^ ~ &` is correct, segment order is valid, and OBX-8 uses HL7 Table 0078 interpretation codes properly.

REMEDY Complete MSH-9, make MSH-3 to MSH-6 configurable per interface, add MSH-18, and emit a batch envelope when more than one message is written.

F-06 · One Health ATC code is required but never validated

Medium

WHERE `one-health-engine.ts:58` — `field('atc_code', 'ATC code', 'text', true)`

OBSERVED A captured antimicrobial-consumption event, read back from the database:

```
{ "antimicrobial": "Meropenem", "atc_code": "J01DH02", "aware_group": "Watch",  
  "quantity_grams": 1450, "who_ddd_grams": 3, "bed_days": 9200, "patient_days": 8800 }
```

The value above is correct, but only because it was typed correctly. The field is a plain text input: no format check, no lookup against the 299 ATC codes already in `master_antibiotics.atc_code`, and no cross-check that the AWaRe group matches the one the catalogue records for that agent.

IMPACT Consumption data keyed on a mistyped ATC code cannot be aggregated nationally, and the error is invisible until reconciliation fails.

REMEDY Turn the field into a lookup over the antibiotic catalogue, auto-populating the AWaRe group and the WHO DDD, with free text allowed only as an explicitly flagged override.

F-07 · One Health observation time is not a valid ISO 8601 instant

Medium

WHERE Capture form `datetime-local` value stored verbatim; emitted at `one-health-exporters.ts:23` and `:63`

OBSERVED

```
observed_at = "2026-08-12T03:31"    <-- no seconds, no timezone offset
```

The companion `recorded_at` is a proper UTC instant. `observed_at` is not, and it is the field carried into the GLASS, ANIMUSE and InFARM products as the observation date.

IMPACT An offset-free local datetime is ambiguous once it crosses a timezone boundary, and strict ISO 8601 or FHIR `dateTime` parsers reject it.

REMEDY Normalise on capture to a full offset-bearing instant (`2026-08-12T03:31:00+05:30`), the same way isolate specimen dates are normalised.

F-08 · Diagnosis is captured but has no ICD binding and is never exported

Medium

WHERE	Schema <code>database.ts:621</code> ; import aliases <code>services.ts:88</code> ; editor field with no system selector; no reference in any export builder
OBSERVED	The database has <code>diagnosis_system</code> , <code>diagnosis_code</code> and <code>diagnosis_display</code> . The import mapper recognises <code>icd</code> and <code>icd code</code> as aliases for <code>diagnosis_code</code> . The isolate editor offers a plain “Diagnosis code” text box with no code-system choice. No FHIR <code>Condition</code> , no <code>DiagnosticReport.conclusionCode</code> and no HL7 <code>DG1</code> segment is produced.
IMPACT	Clinical context that laboratories are already entering is invisible to every receiver. ICD-11 appears in the application only inside an assistant prompt string (<code>services.ts:1168</code>).
REMEDY	Bind <code>diagnosis_system</code> to a small system list (ICD-10, ICD-11 MMS, SNOMED CT), validate the code shape, and emit <code>DiagnosticReport.conclusionCode</code> in FHIR and a <code>DG1</code> segment in HL7.

F-09 · No profile claim, and no ABDM / NDHM alignment

Medium

WHERE	<code>services.ts:825-912</code>
OBSERVED	No resource carries <code>meta.profile</code> ; the bundle has no <code>identifier</code> ; <code>Bundle.type</code> is <code>collection</code> ; the patient identifier namespace is <code>urn:amrit:patient:<LAB></code> ; no <code>Practitioner</code> or <code>PractitionerRole</code> is produced.
IMPACT	Without a profile claim there is nothing for a validator to check against, so “FHIR-compliant” cannot be asserted or tested. For Ayushman Bharat Digital Mission exchange specifically, a health record must be a <code>document</code> Bundle led by a Composition, patients must be identified by ABHA number or address, and resources must claim the relevant NDHM profile — none of which is present.
REMEDY	Publish an ICMR AMRIT implementation guide with profiles for the five resources produced; stamp <code>meta.profile</code> ; add an ABHA identifier slice; and add a <code>document</code> Bundle variant with a Composition for ABDM exchange.

F-10 · GLASS, ANIMUSE and InFARM products are structural, not coded

Medium

WHERE `one-health-exporters.ts:14-70`

OBSERVED Each product declares a self-defined profile string — `WHO-GLASS-compatible/1.0`, `WOAH-ANIMUSE-aligned/1.0`, `FAO-InFARM-compatible/1.0` — and then passes payload values through unchanged:

```
organism:    values.organism    ?? null    // free text
specimen:    values.sample_type ?? null    // free text
susceptibility: values.ast_summary ?? null    // free text
country:     'IND'              // hardcoded (ANIMUSE)
```

IMPACT The naming invites a reader to believe these are submission-ready. They are not: none of the three programmes' code lists, required-field sets or specimen-organism validity rules are applied, and no submission schema is validated against. They are useful staging extracts.

REMEDY Either bind the fields to the programme code lists and validate against the published submission schema, or rename the products to make their staging role explicit. The hardcoded `IND` should come from the laboratory's country field.

F-11 · Duplicate organism concepts weaken name-based canonicalisation

Low

WHERE `master_organisms`; matching logic at `decision-support.ts:379-381`

OBSERVED Both `ECOLI` and `ECO` exist with organism name “Escherichia coli” and the same SNOMED concept `112283007`. Name-based matching selects the first row in sort order, so which working code a record acquires depends on catalogue ordering rather than intent.

IMPACT Two imports of the same organism name can acquire different working codes, fragmenting local analytics.

REMEDY Deduplicate on concept code, mark one row canonical and the other an alias, and make name-based matching deterministic by preferring the canonical row.

F-12 · PBKDF2 iteration count is below current guidance

Low

WHERE `one-health-governance.ts:3` — `PBKDF2_ITERATIONS = 310_000`

IMPACT 310 000 iterations of PBKDF2-HMAC-SHA256 meets the earlier OWASP figure but is below the current recommendation of 600 000. This is a hardening item, not a vulnerability: salts are 16 random bytes, digests are 32 bytes and comparison is constant-time.

REMEDY Raise the constant. The stored format already encodes its own iteration count and the verifier accepts 100 000–2 000 000, so existing hashes keep verifying and new or re-set passwords use the higher value — no migration is required.

F-13 · The WHONET CSV header does not use WHONET field names

Medium

WHERE `services.ts:614-642` — `WHONET_CORE_HEADERS`, consumed at `:737-744`

OBSERVED The header row of a produced `.csv`:

```
Patient ID,Specimen number,Specimen date,Specimen type,Organism,Organism code,Sex,...
```

WHONET identifies its own columns by short field names, not by prose labels. The application ships the dictionary that proves it: `fieldDefinitions` holds 425 distinct WHONET field names across 898 rows and 33 modules, including `PATIENT_ID`, `SPEC_NUM`, `SPEC_DATE`, `SPEC_TYPE`, `ORGANISM`, `SEX`, `DATE_BIRTH`, `WARD`, `DEPARTMENT`, `INSTITUT`, `COUNTRY_A`, `PAT_TYPE`, `DATE_ADMIS` and `AGE`.

IMPACT A national node running WHONET or BacLink must define a column mapping before the file can be loaded, once per exporting site. The content is right; the labelling is not.

REMEDY Emit WHONET field names in the header row, driven from the packaged `fieldDefinitions` table, and offer the prose labels as a separate human-readable export. The antibiotic result columns should follow WHONET's own `<CODE>_ND` / `<CODE>_NM` convention for disk and MIC.

F-14 · Genomic markers carry no database accession

Low

WHERE `resources/genomic-markers.v1.json` — 52 markers

OBSERVED Each marker carries a gene symbol, mechanism class, predicted resistance, linked antibiotic codes, a default method and a reporting standard, but no accession into any reference database:

```
{ "code": "BLANDM", "gene_symbol": "blaNDM",
  "mechanism_class": "Carbapenemase (Ambler class B, MBL)",
  "reporting_standard": "WHO GLASS-WGS; ICMR-AMRSN" }
    <-- no NCBI Reference Gene Catalog accession, no CARD ARO id
```

IMPACT Gene symbols are not stable identifiers: `blaNDM` names a family whose members differ in substrate profile. Without an accession, a genotypic result cannot be reconciled against a sequencing pipeline's own output, which is what GLASS-WGS reporting requires.

POSITIVE The source citations behind the set are complete and specific — WHO GLASS-WGS, the GLASS colistin and GLASS-EAR guidance, the WHO Bacterial Priority Pathogens List 2024, the WHO tuberculosis mutation catalogue and ICMR AMRSN — which is more provenance than most marker lists carry.

REMEDY Add an NCBI Reference Gene Catalog accession and, where one exists, a CARD ARO identifier to each marker, and surface both in the Master Studio editor.

22.1 Findings by severity

SEVERITY	COUNT	FINDINGS
High	3	F-01 mismatched code system · F-02 no LOINC · F-03 no UCUM units
Medium	8	F-04 · F-05 · F-06 · F-07 · F-08 · F-09 · F-10 · F-13

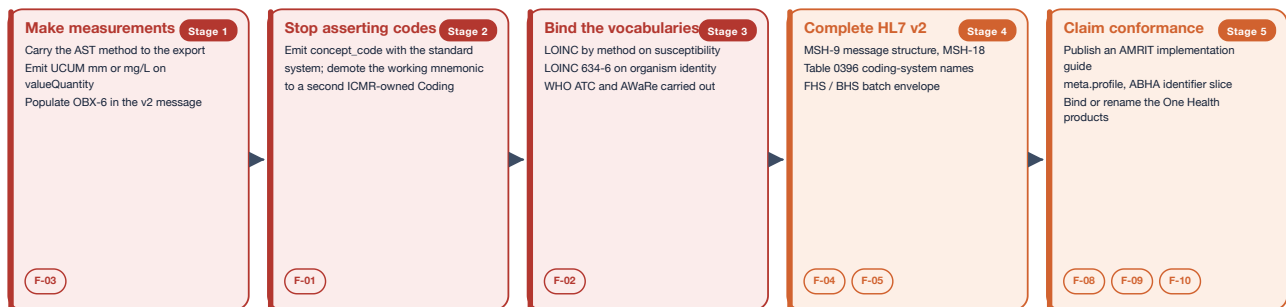
SEVERITY	COUNT	FINDINGS
Low	3	F-11 duplicate concepts · F-12 KDF iterations · F-14 marker accessions

F-13 and F-14 were raised by the re-verification pass of 12 August 2026. Neither changes the high-severity picture; both are recorded because they are cheap to close and each removes a manual step from a national workflow.

23 · Remediation roadmap

The good news in this report is that eleven of the fourteen findings are export-layer changes. The terminology content, the validation discipline and the governance model are already in place; what is missing is carrying them across the boundary.

WHAT HAS TO HAPPEN, IN WHAT ORDER, TO SUPPORT A FULL CONFORMANCE CLAIM



ACCEPTANCE TESTS THAT CLOSE EACH STAGE

Stage 1 Every valueQuantity carries system <http://unitsofmeasure.org> and code mm or mg/L.

Stage 2 No Coding pairs <http://snomed.info/sct> with a code that is not all digits.

Stage 3 Every susceptibility Observation.code has a LOINC coding alongside the local one.

Stage 4 Every emitted CWE third component is a Table 0396 value or a 99-prefixed namespace.

Stage 5 The Bundle validates against the published AMRIT implementation guide, zero errors.

Stages 1 and 2 are the only ones that change what a clinician or an analyst can conclude from a received record.
They are also the cheapest: both are contained inside the exporter and the canonicaliser, with no schema migration.

Figure 16 — Remediation roadmap. The five grouped stages, the findings each one closes, and the acceptance test that closes it. Stages 1 and 2 are the only ones that change what a clinician or an analyst can conclude from a received record; they are also the cheapest, being contained inside the exporter and the canonicaliser with no schema migration.

23.1 Sequenced work

STAGE	CHANGE	RESOLVES	WHY IN THIS ORDER
1	Attach UCUM units and the test method to every quantitative AST value, in FHIR and in OBX-6	F-03	Highest consequence, smallest surface, no data migration — the method is already stored beside the measurement.
2	Emit <code>concept_code</code> with its standard system and demote the working code to a second Coding under an ICMR CodeSystem URI	F-01	Stops the application asserting codes that do not exist. Read-only change to the export builders.
3	Bind observation codes to LOINC using <code>loinc_mlc</code> / <code>loinc_sbt</code> by method; LOINC 634-6 for organism identification	F-02	Makes results findable. Uses catalogue content already present for 121 agents; extend coverage incrementally.

STAGE	CHANGE	RESOLVES	WHY IN THIS ORDER
4	Complete the HL7 v2.5.1 header, map coding systems to Table 0396, add a batch envelope	F-04, F-05	Independent of the FHIR work; unblocks strict interface engines.
5	Normalise One Health <code>observed_at</code> ; make ATC a validated lookup that populates AWaRe and DDD	F-06, F-07	Small capture-form changes with a direct effect on national consumption reporting.
6	Bind <code>diagnosis_system</code> to a system list and export <code>conclusionCode / DG1</code>	F-08	Turns already-captured clinical context into exportable content.
7	Publish an AMRIT implementation guide; stamp <code>meta.profile</code> ; add an ABHA identifier slice and a document-Bundle variant	F-09	Once codes are correct, a conformance claim becomes testable rather than aspirational.
8	Bind the One Health products to programme code lists, or rename them as staging extracts	F-10	Depends on the programme submission specifications being obtained.
9	Emit WHONET field names in the CSV header, driven from the packaged field dictionary	F-13	Removes a per-site mapping step for every WHONET-based national node. Header-row change only.
10	Deduplicate organism concepts; add marker accessions; raise the PBKDF2 iteration count	F-11, F-12, F-14	Housekeeping; none of the three blocks the others.

23.2 Suggested acceptance tests

Each remediation should land with a regression test that asserts the standard directly, so conformance cannot silently regress:

- Every `valueQuantity` in an exported bundle has `system = http://unitsofmeasure.org` and a unit code in { `mg/L` , `mm` }.
- No `Coding` pairs `http://snomed.info/sct` with a code that is not all digits.
- Every susceptibility `Observation.code` contains a coding with `system = http://loinc.org`.
- Every emitted HL7 CWE third component is a Table 0396 value or a `99` -prefixed local namespace.
- Every One Health `observed_at` matches a full ISO 8601 instant with an offset.
- A produced bundle validates against the published AMRIT implementation guide.
- Every column in a produced WHONET CSV header matches a `field_name` in the packaged field dictionary.

23.3 Effort and risk

STAGE	SURFACE	DATA MIGRATION	RISK
1 – 4	Export builders only	None	Low. Existing exports change shape, so downstream consumers must be told.
5 – 6	Capture forms + schema binding	Optional back-fill of <code>observed_at</code>	Low.
7	New implementation guide and profiles	None	Medium — requires a published, versioned IG to be maintained.
8	Programme mappings	None	Medium — depends on external specifications.
9	Export header row only	None	Low. Consumers of the current header must be told.
10	Catalogue and one constant	Catalogue dedup only	Low.

24 · Statement of conformance

Verdict

As built at version 2.0.0, ICMR AMRIT is **substantially conformant and partially interoperable**. It cannot be described as fully compliant with national and international data terminology standards, nor as fully interoperable, and the three high-severity findings must be closed before such a claim could be made.

24.1 What is demonstrably sound

- **Structural FHIR R4.** Collection bundles with UUID-URN `fullUrl` values and deterministic RFC 4122 version-5 resource identifiers, asserted in code and verified in produced artefacts. Internal references resolve.
- **Correct use of HL7 terminology.** `v3-observationInterpretation` for S/I/R, `observation-category`, `v2-0074` for the report class, and `measure-scoring` / `measure-population` in the MeasureReport.
- **UCUM where it is used.** The MeasureReport percentage carries `system: http://unitsofmeasure.org` and code `%`.
- **A serious terminology catalogue.** 88.2 % SNOMED CT organism coverage, 86.9 % GBIF taxonomy, 74.9 % WHO ATC including 31 ATCvet codes for the veterinary agents, WHO AWaRe 2025 with its source citation, LOINC susceptibility codes, 2 710 coded values across twelve sets, and a 425-name WHONET field dictionary.
- **National geography done properly.** LGD codes for all 36 states and union territories and 785 districts, bound in laboratory setup and One Health capture, with ISO 3166-1 alpha-3 countries carrying alpha-2 and WHO region.
- **Closed-vocabulary enforcement.** Unknown or inactive codes, and contradictory name/code pairs, are hard errors — verified by observing the save path reject them.
- **Content integrity.** The packaged catalogue is size-checked, schema-checked, row-count-checked, referentially checked, PII-scanned and SHA-256 pinned to the reviewed manifest.
- **Governance.** Six-role RBAC, PBKDF2 identities, memory-only sessions, a hash-chained audit log with a language-independent canonical form, explicit breakpoint activation, dialog-approved file destinations and a doubly guarded aggregate-only federation path.

24.2 What prevents a full-compliance claim

1. Codes are emitted under a code system they do not belong to (F-01).
2. No LOINC identifies any observation (F-02).
3. Quantitative susceptibility values carry no unit, so MIC and zone diameter are indistinguishable to a receiver (F-03).
4. HL7 v2 messages use invalid coding-system names and an incomplete header, with no batch envelope (F-04, F-05).
5. WHO ATC and AWaRe, held for hundreds of agents, never reach an export; the One Health ATC field is unvalidated (F-02, F-06).
6. There is no ICD binding, no profile claim, and no ABDM/NDHM alignment (F-08, F-09).
7. The GLASS, ANIMUSE and InFARM products are structurally shaped but content-uncoded (F-10).

8. The WHONET CSV is content-complete but its header does not use WHONET field names, so it needs a mapping step at every receiving node (F-13).

24.3 A defensible interim claim

Wording that the evidence supports today

“ICMR AMRIT 2.0.0 produces structurally valid HL7 FHIR R4 and HL7 v2.5.1 artefacts and WHONET-compatible extracts from a SNOMED CT-, WHO ATC/ATCvet-, WHO AWaRe- and LGD-coded master catalogue. Terminology binding in the exported artefacts is partial: LOINC identification and UCUM units are not yet emitted, and no implementation-guide conformance is claimed. Federation is aggregate-only and no patient-level record leaves the device.”

Once stages 1 to 4 of the roadmap are complete and the acceptance tests in §23.2 pass, the claim can be strengthened to full terminology binding for the laboratory workflow. An ABDM/NDHM claim additionally requires stage 7.

24.4 Reviewer sign-off

ROLE	NAME	DATE AND SIGNATURE
Laboratory authority (breakpoints)		
Data standards reviewer		
Programme owner		

Appendix A · Export field reference

A.1 FHIR R4 Bundle

RESOURCE	ELEMENT	SOURCE AND CURRENT CONTENT
Organization	identifier[0]	urn:whonet:lab-code + laboratory code
	identifier[1]	urn:whonet:site-group + site group, when set
	address[0]	country, state, district from the laboratory record
Patient	identifier	urn:amrit:patient:<LAB> + patient identifier
	gender	male / female / unknown
	birthDate	normalised date
Specimen	identifier	urn:amrit:specimen:<LAB> + specimen number
	type.coding	specimen code with specimen_system, defaulting to urn:whonet:specimen-code F-01
	collection.collectedDateTime	specimen date at T00:00:00Z
Observation (organism)	code	text “Organism identified” — no coding F-02
	valueCodeableConcept	organism code with organism_system F-01
	category	observation-category = laboratory ok
Observation (AST)	code.coding	urn:whonet:antibiotic-code F-02
	interpretation	v3-ObservationInterpretation R/I/S ok
	valueQuantity	bare number, no unit F-03
DiagnosticReport	status	preliminary for drafts, final otherwise ok
	category	v2-0074 = MB, Microbiology ok
	code	free text F-02

A.2 HL7 v2.5.1 segments

SEGMENT	CONTENT
MSH	Sending app AMRIT, sending facility = laboratory code, ORU^R01, processing P, version 2.5.1 F-05
PID	Patient identifier, name, date of birth, sex M/F/U
PV1	Patient class I or O derived from location type; location and department
SPM	Specimen id, specimen code ^ type ^ system, collection date F-04
OBR	MICR0^Microbiology Culture and Susceptibility^L, specimen date, performing laboratory

SEGMENT	CONTENT
OBX 1	CWE organism identification F-02, F-04
OBX 2+	One per antibiotic, sorted by code. NM when numeric, CWE when interpretation only, ST otherwise. OBX-6 units empty F-03
NTE	AST-not-performed reason, infection origin, alert summary

A.3 WHONET CSV columns

27 core columns followed by, for each antibiotic in the cohort, a result column and a measurement column, then origin, alerts and expert comments, then per-antibiotic method, guideline, potency and source columns. Headers are human-readable labels (“Patient ID”, “Specimen number”) rather than WHONET field names, so a receiving WHONET installation needs a one-time column mapping.

A.4 FHIR MeasureReport

A proportion Measure with canonical URL `urn:whonet:measure:amr-resistance-rate:<ABX>`, populations for initial-population, denominator and numerator, a measure score as a UCUM percentage, and a stratifier by organism. The denominator is S + I + R and the numerator is R. Drafts are excluded. This is the most standards-complete artefact the application produces.

Appendix B · Reproducing the verification

B.1 Build and test

```
cd electron_app
npx electron-vite build
npx vitest run          # expect 100 passed, 2 skipped
```

B.2 Run against an isolated copy of the data

Never verify against the operator's live database. Copy it — including the `-wal` and `-shm` files — and point the application at the copy with its own user-data directory, so the running production instance is undisturbed and the single-instance lock does not reject the verification run:

```
cp "$HOME/Library/Application Support/ICMR AMRIT/data/amrit.sqlite3"* "$SANDBOX/"

AMRIT_DATABASE_PATH="$SANDBOX/amrit.sqlite3" \
./node_modules/.bin/electron . \
--user-data-dir="$SANDBOX/userdata" \
--remote-debugging-port=9223
```

B.3 Measure catalogue coverage

```
python3 - <<'PY'
import json
c = json.load(open('resources/catalog-seed.v1.json'))['catalogue']
def cov(rows, cols):
    for col in cols:
        n = sum(1 for r in rows if str(r.get(col) or '').strip())
        print(f'{col:22} {n:5}/{len(rows)} {100*n/len(rows):5.1f}%')
cov(c['organisms'], ['snomed_code', 'concept_code', 'system', 'gbif_taxon_id'])
cov(c['antibiotics'], ['atc_code', 'loinc_mlc', 'loinc_sbt', 'who_aware'])
PY
```

B.4 Produce export artefacts

Add a temporary Vitest file that seeds a database from the packaged catalogue, saves one isolate through `AMRITDatabase.saveRecord` and calls `createExport` for each format:

```
const database = new AMRITDatabase(path, {
  seedCatalog: true,
  catalogSeedPath: join(process.cwd(), 'resources', 'catalog-seed.v1.json')
}).initialize()
database.saveLab({ code: 'LAB-1', name: '...', state_lgd_code: '32', ... })
database.selectLab('LAB-1')
database.saveRecord({ ... specimen_type: 'Urine', organism: 'Escherichia coli',
  antibiotic_results: { CIP: { result: 'R', measurement: 4,
    method: 'MIC' } } })
for (const format of ['fhir', 'hl7', 'whonet'])
  writeFileSync(`out.${format}`, createExport(format, rows, lab, {}))
```

Delete the temporary file afterwards so it does not join the permanent suite. The findings in Chapter 22 quote the artefacts this produces verbatim.

B.5 Count code-system agreement directly

The scale of Finding F-01 is a SQL question, not an inspection question. Run it against the sandboxed copy:

```
-- master rows that will produce a false SNOMED CT assertion
SELECT COUNT(*) FROM master_organisms
  WHERE system='http://snomed.info/sct' AND code NOT GLOB '[0-9]*';
SELECT COUNT(*) FROM master_samples
  WHERE system='http://snomed.info/sct' AND code NOT GLOB '[0-9]*';

-- how the stored isolates are actually coded today
SELECT organism_system,
       CASE WHEN organism_code GLOB '[0-9]*' THEN 'numeric' ELSE 'alphanumeric' END,
       COUNT(*) FROM isolates GROUP BY 1,2;
```

B.6 Drive the compiled interface

Attach to the Chrome DevTools Protocol endpoint on the debugging port. Two behaviours matter: the `window.amrit` bridge is frozen and cannot be patched, so call it directly; and Electron refuses the websocket with `403 Forbidden` unless the `Origin` header is suppressed. Set React-controlled inputs through the native value setter followed by an `input` event, or the component will not observe the change.

B.7 Verification artefacts produced for this report

ARTEFACT	PURPOSE
masters.json	Catalogue rows for the concepts under test
export.fhir / hl7 / whonet	Synthetic-path artefacts from a freshly seeded catalogue
real.fhir / hl7 / whonet	Live-path artefacts from the operator database snapshot
export.measure.json	MeasureReport for a single antibiotic
aggregate.json	Federated organism and specimen distribution codings
saved-record.json	The record as canonicalisation rewrote it
coding-report.json	Direct SQL counts of code / system agreement across the whole isolate table and both master catalogues — the evidence behind the <i>Scale</i> row of F-01
Screens 1–29	Compiled build driven over CDP against the sandboxed database, including an authenticated One Health session and a governed capture

Appendix C · Glossary and abbreviations

TERM	MEANING
ABDM / NDHM	Ayushman Bharat Digital Mission, India's national digital health architecture; formerly the National Digital Health Mission.
ABHA	Ayushman Bharat Health Account — the citizen health identifier used by ABDM.
AMRSN	ICMR's Antimicrobial Resistance Surveillance and Research Network.
AST	Antimicrobial susceptibility testing.
ATC	WHO Anatomical Therapeutic Chemical classification of medicinal substances.
AWaRe	WHO Access / Watch / Reserve classification of antibiotics; 2025 edition used here.
BacLink	WHONET's data-import companion for laboratory instrument and LIS files.
CDP	Chrome DevTools Protocol — used here to drive and photograph the compiled build.
CLSI / EUCAST	The two breakpoint standards bodies AMRIT supports.
CWE	HL7 v2 “coded with exceptions” data type: code ^ text ^ coding-system name.
DDD	WHO Defined Daily Dose — the consumption denominator.
FHIR	HL7 Fast Healthcare Interoperability Resources; release 4 is used.
GBIF	Global Biodiversity Information Facility — the taxonomy backbone referenced by the organism catalogue.
GLASS	WHO Global Antimicrobial Resistance and Use Surveillance System.
HAI	Healthcare-associated infection (CLABSI, CAUTI, VAP, SSI).
InFARM	FAO's International FAO Antimicrobial Resistance Monitoring system.
IPC	Infection prevention and control.
LGD	Local Government Directory — India's authoritative state and district code list.
LOINC	Logical Observation Identifiers Names and Codes — identifies what a laboratory test measures.
MIC	Minimum inhibitory concentration, reported in mg/L.
ORU^R01	The HL7 v2 unsolicited observation-result message.
PBKDF2	Password-Based Key Derivation Function 2, used with HMAC-SHA256 for local identities.
PCU	Population correction unit — the animal-biomass denominator for veterinary antimicrobial use.
PHI	Protected health information.
S / I / R	Susceptible, intermediate, resistant.
SNOMED CT	Systematized Nomenclature of Medicine Clinical Terms — organism and specimen concepts.
UCUM	Unified Code for Units of Measure — the unit system FHIR and HL7 expect.
WAL	SQLite write-ahead logging journal mode.

TERM	MEANING
WHONET	WHO's free AMR surveillance software and its code space for organisms and antibiotics.
WOAH ANIMUSE	World Organisation for Animal Health's antimicrobial-use database.
zod	The TypeScript schema library used to validate every IPC payload.

Document control

Title	ICMR AMRIT 2.0.0 — Application Manual and Standards Conformance Report
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Build verified	electron-vite build of the working tree; catalogue 2026.1
Evidence base	Source inspection, catalogue measurement, produced export artefacts, and interactive verification of the compiled build
Status	Substantially conformant; three high-severity findings open
Next review	On completion of roadmap stages 1–4, or on the next catalogue version